

Well Integrity Report - Form 19 Summary

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840

Well File No. **90329**

UIC No. **W0437S0919D**

Test Date: **3/31/2020 10:58:29 AM** Contractor Performing Pressure Test: **Castle**
Operator: **OASIS PETROLEUM NORTH AMERICA LLC** Telephone : **2814049500**
Address: **1001 FANNIN STE 1500** City: **HOUSTON** State: **TX** Zip Code: **77002**
Well Name: **CARSON SWD 5301 12-24** Field: **BAKER**
Location: Qtr-Qtr: **NENW** Section: **24** Township: **153** Range: **101** Cnty: **MCKENZIE**

WELL TEST DATA

Formation:	DAKOTA		Perforations:	5950-7080	<i>Feet</i>
Tool Type:	PKR	Depth:	5821	Packer Model:	Nickle plated Arrowset
Tubing Size:	4.5	<i>Inches</i>	Tubing Type:	11.6 # J-55 Polycore	
Well Type:	Disposal Well	Reason for Test:	UIC MIT	Test Type :	5 Year MIT
BEFORE TEST START of TEST END of TEST					
Tubing Pressure:	1400	<i>PSIG</i>	Flowing	1400	<i>PSIG</i> Flowing
Annulus or Casing Pressure:	0	<i>PSIG</i>		1000	<i>PSIG</i>
Annulus or Casing Fluid :	Treated Salt Water		Test Fluid:	Salt Water	
Amount of Fluid Needed to Fill Annulus or Casing:	0.25		<i>Bbls</i>		
Test Length:	15	<i>Minutes</i>	Was Annulus or Casing Bled to Zero After Test?:	Yes	If No - Pressure Left: ☐

INSPECTION COMMENTS

Test Result:	Acceptable	Report of Work Done Required:	No
MIT passed with no pressure loss over 15 min, MIT due for 5 year test.			

This Report is true and complete to the best of my knowledge.

Company Representative Witnessing Test:

Joe Brock

Title:

SWD Field Foreman

Commission Field Inspector Witnessing Test:

William Ryan

NDIC Initials

JS



AUTHORIZATION TO PURCHASE AND TRANSPORT OIL FROM LEASE – Form 8
INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5698 (03-2000)

RECEIVED

APR - 8 2019

ND OIL & GAS DIVISION

Well File No.

90329

NDIC CTB No.

650209

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND FOUR COPIES.

Well Name and Number CARSON SWD 5301 12-24	Qtr-Qtr NENW	Section 24	Township 153	Range 101	County Mckenzie
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Operator Oasis Petroleum North America LLC	Telephone Number (281) 404-9573	Field BAKER
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Address 1001 Fannin, Suite 1500	City Houston	State TX	Zip Code 77002
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Name of First Purchaser Oasis Petroleum Marketing LLC	Telephone Number (281) 404-9627	% Purchased 100%	Date Effective January 1, 2019
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Principal Place of Business 1001 Fannin, Suite 1500	City Houston	State TX	Zip Code 77002
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Field Address	City	State	Zip Code
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Transporter Power Energy Logistics, LLC	Telephone Number (512) 363-5296	% Transported 100%	Date Effective January 1, 2019
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Address 1927 Lohmans Crossing Suite 103	City Lakeway	State TX	Zip Code 78734
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The above named producer authorizes the above named purchaser to purchase the percentage of oil stated above which is produced from the lease designated above until further notice. The oil will be transported by the above named transporter.

Other Transporters Transporting From This Lease	% Transported	Date Effective
		January 1, 2019
Other Transporters Transporting From This Lease	% Transported	Date Effective
		January 1, 2019
Other Transporters Transporting From This Lease	% Transported	Date Effective
		January 1, 2019
Other Transporters Transporting From This Lease	% Transported	Date Effective
		January 1, 2019
Comments		

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.	Date April 3, 2019
Signature 	Printed Name Claudia Arguelles
	Title Contracts Administrator

Above Signature Witnessed By:	
Signature 	Printed Name Kenzie Buchanan
	Title Scheduler

FOR STATE USE ONLY

Date Approved APR 11 2019
By Melissa Herz
Title Oil & Gas Production Analyst



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.dmr.nd.gov/oilgas/

90329

May 11, 2016

RE: Saltwater Disposal Well Reporting and Monitoring Requirements

Dear Saltwater Disposal Operator:

This letter is to outline and clarify the Commission's requirements for monthly reporting and monitoring of saltwater disposal wells which are outlined under North Dakota Administrative Code (NDAC) Section 43-02-05-12, and the type of wastes that can be injected pursuant to the EPA guidelines for Exempt and Non-Exempt Oil and Gas Exploration and Production Wastes.

All injection into a saltwater disposal well must be measured, either through a meter or by an approved method. Accurate gauges must be placed on the tubing and tubing-casing annulus of the well for pressure monitoring. Any noncompliance with regulations or permit conditions must be reported to the Commission orally within 24 hours followed by a written explanation within 5 days.

Monthly reporting of the following information is required to be submitted for a saltwater disposal well on a Saltwater Disposal Report (Form 16):

1. Beginning and end of month meter readings.
2. Total barrels injected for the month.
3. Average daily operating injection pressure (this means a typical pressure observed while the well is injecting, not a mathematical average).
4. Sources of received fluid (these sources need to be listed by well or facility name and number, NDIC well or facility file number, and the location of each source by quarter-quarter, section, township, and range).
5. Amount of fluid received from each source.
6. Nature of the received fluid (i.e. produced water, pit water, etc.).
7. Transportation method used to transport the injection fluid to the disposal well site (i.e. pipeline or truck).

Saltwater Disposal Reports (Form 16) are due on or before the fifth day of the second month succeeding the month in which the well is capable of disposal (i.e. January report due March 5th).

Pursuant to NDAC Section 43-02-05-13.1, all records from which to make and substantiate required reports shall be kept for a period of not less than six years.

May 11, 2016

Saltwater Disposal Well Reporting and Monitoring Requirements

Page 2 of 2

The NDIC abides by the Environmental Protection Agency guidelines for Exempt and Non-Exempt Oil and Gas Exploration and Production Wastes to determine what wastes can be accepted for disposal from oil and gas exploration and production operations. Attached is an example list of exempt and non-exempt oil and gas exploration and production wastes along with a flow chart for determining the status of a waste.

The operator is only authorized to inject Class II fluids into a saltwater disposal well. Class II fluids are generally defined as follows:

1. Produced water from oil and gas production.
2. Waste fluids from actual drilling operations.
3. Used workover, completion, and stimulation fluids recovered from production, injection, and exploratory wells.
4. Waste fluids from circulation during well cementing.
5. Pigging fluids from cleaning of collection and injection lines within the field.
6. Enhanced recovery waters including fresh water makeup and other waters containing chemicals for the purpose of enhanced recovery.
7. Gases used for enhanced recovery/pressure maintenance of production reservoirs.
8. Brine reject from water softeners associated with enhanced recovery.
9. Waste oil and fluids from clean up within the oil field.
10. Waste water from gas plants which are an integral part of production operations unless those waters are classified as hazardous waste at the time of injection.

The following fluids are non-exempt fluids and are not authorized to be injected into a Class II saltwater disposal well:

1. Unused fracturing fluids or acids.
2. Vacuum truck and drum rinsate from trucks and drums containing non-exempt waste.
3. Refined gasoline or diesel fuels.
4. Used equipment lubricating oils.
5. Refinery wastes.
6. Radioactive tracer wastes.
7. Car wash wastes.

Please distribute this letter and the attachment to the appropriate personnel responsible for reporting on and monitoring each saltwater disposal well your company operates. If you have any questions, do not hesitate to contact me.

Sincerely,

Ashleigh Day
Underground Injection Supervisor

**WELL COMPLETION OR RECOMPLETION REPORT - FORM 6**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 2468 (04-2010)



Well File No. **90329**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.

PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Designate Type of Completion					
☐ Oil Well	☐ EOR Well	☐ Recompletion	☐ Deepened Well	☐ Added Horizontal Leg	☐ Extended Horizontal Leg
☐ Gas Well	☒ SWD Well	☐ Water Supply Well	☐ Other:		
Well Name and Number Carson SWD 5301 12-24			Spacing Unit Description		
Operator Oasis Petroleum North America		Telephone Number (281) 404-9591	Field Baker		
Address 1001 Fannin, Suite 1500			Pool Dakota		
City Houston	State TX	Zip Code 77002	Permit Type ☐ Wildcat ☒ Development ☐ Extension		

LOCATION OF WELL

At Surface 584 F N L	2477 F WL	Qtr-Qtr NENW	Section 24	Township 153 N	Range 101 W	County McKenzie
Spud Date April 13, 2015	Date TD Reached April 19, 2015	Drilling Contractor and Rig Number Nabors B27		KB Elevation (Ft) 2109	Graded Elevation (Ft) 2084	
Type of Electric and Other Logs Run (See Instructions) MWD/GR from surface to TD						

CASING & TUBULARS RECORD (Report all strings set in well)

Well Bore	String Type	Size (Inch)	Top Set (MD Ft)	Depth Set (MD Ft)	Hole Size (Inch)	Weight (Lbs/Ft)	Anchor Set (MD Ft)	Packer Set (MD Ft)	Sacks Cement	Top of Cement
Surface Hole	Surface	9 5/8	0	2210	13 1/2	36			812	0
Vertical Hole	Intermediate	7	0	7193	8 3/4	32			835	2180

PERFORATION & OPEN HOLE INTERVALS

Well Bore	Well Bore TD Drillers Depth (MD Ft)	Completion Type	Open Hole/Perforated Interval (MD Ft)		Kick-off Point (MD Ft)	Top of Casing Window (MD Ft)	Date Perf'd or Drilled	Date Isolated	Isolation Method	Sacks Cement
			Top	Bottom						
Lateral 1	7217	Perforations	5950	7080	4695		05/01/2015			

PRODUCTION

Current Producing Open Hole or Perforated Interval(s), This Completion, Top and Bottom, (MD Ft) Lateral 1-						Name of Zone (If Different from Pool Name)			
Date Well Completed (SEE INSTRUCTIONS) June 18, 2015		Producing Method		Pumping-Size & Type of Pump			Well Status (Producing or Shut-In)		
Date of Test	Hours Tested	Choke Size /64	Production for Test	Oil (Bbls)	Gas (MCF)	Water (Bbls)	Oil Gravity-API (Corr.)	Disposition of Gas	
Flowing Tubing Pressure (PSI)		Flowing Casing Pressure (PSI)		Calculated 24-Hour Rate	Oil (Bbls)	Gas (MCF)	Water (Bbls)	Gas-Oil Ratio	

GEOLOGICAL MARKERS

[illegible]

PLUG BACK INFORMATION

[illegible]

CORES CUT

Top (Ft)	Bottom (Ft)	Formation	Top (Ft)	Bottom (Ft)	Formation

Drill Stem Test

Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
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Sample Chamber Recovery								
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Sample Chamber Recovery								
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Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								

Well Specific Stimulations

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units Barrels
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

ADDITIONAL INFORMATION AND/OR LIST OF ATTACHMENTS

This report has been amended to correct the MD of the Dakota top.

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.	Email Address	Date
	jswenson@oasispetroleum.com	09/23/2015
Signature	Printed Name	Title
	Jennifer Swenson	Regulatory Specialist



RYAN DIRECTIONAL SERVICES, INC.
A NABORS COMPANY

19510 Oil Center Blvd
Houston, TX 77073
Bus 281.443.1414
Fax 281.443.1676

Monday, April 27, 2015

State of North Dakota

Subject: **Surveys**

Re: **Oasis**
Carson SWD 5301 12-24
McKenzie, ND

Enclosed, please find the original and one copy of the survey performed on the above-referenced well by Ryan Directional Services, Inc.. Other information required by your office is as follows:

<i>Surveyor Name</i>	<i>Surveyor Title</i>	<i>Borehole Number</i>	<i>Start Depth</i>	<i>End Depth</i>	<i>Start Date</i>	<i>End Date</i>	<i>Type of</i>	<i>TD Straight Line Projection</i>
Daniel Ogden	MWD Operator	O.H.	2210'	7163'	04/15/15	04/18/15	MWD	7217'

If any other information is required please contact the undersigned at the letterhead address or phone number.

Douglas Hudson
Well Planner



RYAN DIRECTIONAL SERVICES, INC.

A NABORS COMPANY

Ryan Directional Services, Inc.
19510 Oil Center Blvd.
Houston, Texas 77073
Bus: 281.443.1414
Fax: 281.443.1676

Saturday, April 18, 2015

State of North Dakota
County of McKenzie

Subject: **Survey Certification Letter**

Survey Company: **Ryan Directional Services, Inc.**

Job Number: **8858**

Survey Job Type: **Ryan MWD**

Customer: **Oasis Petroleum**

Well Name: **Carson SWD 5301 12-24**

Rig Name: **Nabors B27**

Surface: **404279.54 N, 1206936.03 E**

A.P.I. No: **33-053-90329**

Location: **McKenzie, North Dakota**

RKB Height: **2109'**

Distance to Bit: **54'**

<i>Surveyor Name</i>	<i>Surveyor Title</i>	<i>Borehole Number</i>	<i>Start Depth</i>	<i>End Depth</i>	<i>Start Date</i>	<i>End Date</i>	<i>Type of</i>	<i>TD Straight Line Projection</i>
Daniel Ogden	MWD Supervisor	OH	2273'	7163'	04/14/15	04/18/15	MWD	7217'

The data and calculations for this survey have been checked by me and conform to the calibration standards and operational procedures set forth by Ryan Directional Services, Inc. I am authorized and qualified to review the data, calculations and these reports; the reports represents true and correct Directional Surveys of this well based on the original data, the minimum curvature method, corrected to True North and obtained at the well site.

Daniel Ogden
MWD Supervisor
Ryan Directional Services, Inc.

Report #: **1**
Date: **18-Apr-15**



RYAN DIRECTIONAL SERVICES, INC.
A NABORS COMPANY

Ryan Job # **8858**

SURVEY REPORT

Customer: **Oasis Petroleum**
Well Name: **Carson SWD 5301 12-24**
Rig #: **Nabors B27**
API #: **33-053-90329**
Calculation Method: **Minimum Curvature Calculation**

MWD Operator: **D. Foley / D. Ogden**
Directional Drillers: **RPM**
Survey Corrected To: **True North**
Vertical Section Direction: **180**
Total Correction: **8.20**
Temperature Forecasting Model (Chart Only): **Logarithmic**

Survey #	MD	Inc	Azm	Temp	TVD	VS	N/S	E/W	DLS
Tie in to Gyro Surveys									
Tie In	2210	0.00	0.00	0.00	2210.00	0.00	0.00	0.00	0.00
1	2273	0.80	270.10	84.00	2273.00	0.00	0.00	-0.44	1.27
2	2367	1.10	265.20	91.00	2366.98	0.07	-0.07	-2.00	0.33
3	2461	1.20	260.70	91.00	2460.97	0.31	-0.31	-3.87	0.14
4	2556	1.40	256.40	91.00	2555.94	0.74	-0.74	-5.98	0.23
5	2650	1.70	261.60	95.00	2649.91	1.22	-1.22	-8.47	0.35
6	2744	2.20	264.10	96.00	2743.85	1.60	-1.60	-11.64	0.54
7	2838	0.60	208.20	102.00	2837.82	2.22	-2.22	-13.67	2.05
8	2932	1.10	175.00	104.00	2931.81	3.56	-3.56	-13.83	0.73
9	3026	1.00	182.80	105.00	3025.80	5.27	-5.27	-13.79	0.19
10	3121	0.80	205.50	107.00	3120.79	6.70	-6.70	-14.11	0.43
11	3215	0.80	203.10	107.00	3214.78	7.90	-7.90	-14.65	0.04
12	3309	1.00	211.40	109.00	3308.77	9.20	-9.20	-15.34	0.25
13	3404	1.00	220.10	111.00	3403.75	10.54	-10.54	-16.30	0.16
14	3498	0.40	256.10	113.00	3497.74	11.25	-11.25	-17.15	0.76
15	3592	0.40	271.50	114.00	3591.74	11.32	-11.32	-17.80	0.11
16	3686	0.60	263.90	116.00	3685.74	11.36	-11.36	-18.62	0.22
17	3780	0.30	179.40	118.00	3779.74	11.66	-11.66	-19.10	0.69
18	3874	0.40	186.70	120.00	3873.73	12.23	-12.23	-19.14	0.12
19	3969	0.40	189.50	122.00	3968.73	12.89	-12.89	-19.23	0.02
20	4063	0.30	6.60	123.00	4062.73	12.97	-12.97	-19.26	0.74
21	4157	0.50	12.00	123.00	4156.73	12.32	-12.32	-19.14	0.22
22	4251	0.40	325.50	125.00	4250.73	11.65	-11.65	-19.24	0.39
23	4346	0.40	279.60	127.00	4345.72	11.32	-11.32	-19.76	0.33
24	4440	0.60	265.70	129.00	4439.72	11.31	-11.31	-20.57	0.25
25	4534	0.70	265.40	125.00	4533.71	11.39	-11.39	-21.64	0.11
26	4628	0.90	260.00	127.00	4627.71	11.56	-11.56	-22.94	0.23
27	4646	0.90	267.50	127.00	4645.70	11.59	-11.59	-23.22	0.65
28	4665	0.80	270.70	105.00	4664.70	11.60	-11.60	-23.50	0.58
29	4696	1.50	213.00	105.00	4695.70	11.94	-11.94	-23.94	4.09
30	4727	4.50	188.80	105.00	4726.65	13.48	-13.48	-24.34	10.30
31	4759	9.40	184.20	111.00	4758.41	17.33	-17.33	-24.73	15.40
32	4790	12.50	183.90	114.00	4788.84	23.20	-23.20	-25.14	10.00
33	4821	13.00	183.30	116.00	4819.07	30.03	-30.03	-25.57	1.67
34	4853	13.80	178.30	118.00	4850.20	37.44	-37.44	-25.66	4.40
35	4884	14.10	175.90	120.00	4880.29	44.90	-44.90	-25.28	2.10
36	4915	14.30	174.10	122.00	4910.34	52.47	-52.47	-24.62	1.56
37	4946	14.50	173.70	122.00	4940.37	60.14	-60.14	-23.80	0.72
38	4978	16.40	178.90	123.00	4971.21	68.64	-68.64	-23.27	7.34
39	5009	19.50	184.60	123.00	5000.70	78.17	-78.17	-23.60	11.48
40	5041	22.50	187.40	125.00	5030.57	89.57	-89.57	-24.82	9.88
41	5072	25.40	186.70	129.00	5058.90	102.06	-102.06	-26.36	9.40
42	5103	28.30	184.70	129.00	5086.56	115.99	-115.99	-27.74	9.80
43	5135	31.40	183.80	131.00	5114.31	131.87	-131.87	-28.91	9.79
44	5166	34.10	182.80	131.00	5140.38	148.61	-148.61	-29.87	8.88
45	5197	36.40	181.90	131.00	5165.69	166.49	-166.49	-30.60	7.61
46	5229	36.60	181.30	131.00	5191.42	185.52	-185.52	-31.13	1.28
47	5260	38.60	180.50	131.00	5215.98	204.43	-204.43	-31.43	6.64
48	5292	42.70	179.20	129.00	5240.25	225.27	-225.27	-31.36	13.08
49	5323	44.70	178.60	129.00	5262.66	246.68	-246.68	-30.95	6.59
50	5355	46.90	178.90	129.00	5284.97	269.61	-269.61	-30.45	6.91
51	5386	49.50	178.40	129.00	5305.63	292.71	-292.71	-29.91	8.47
52	5417	52.20	178.40	129.00	5325.20	316.74	-316.74	-29.23	8.71
53	5449	53.00	178.50	131.00	5344.64	342.15	-342.15	-28.55	2.51
54	5480	52.60	178.80	129.00	5363.38	366.84	-366.84	-27.96	1.50
55	5512	54.30	178.40	129.00	5382.43	392.54	-392.54	-27.34	5.41
56	5543	57.10	178.00	129.00	5399.90	418.13	-418.13	-26.53	9.09
57	5575	60.40	178.40	129.00	5416.50	445.47	-445.47	-25.67	10.37
58	5606	64.10	179.80	129.00	5430.93	472.90	-472.90	-25.25	12.59
59	5637	67.70	181.30	129.00	5443.59	501.19	-501.19	-25.52	12.42
60	5669	70.90	181.80	129.00	5454.90	531.11	-531.11	-26.34	10.11

Report #: **1**Date: **18-Apr-15****RYAN DIRECTIONAL SERVICES, INC.**

A NABORS COMPANY

Ryan Job # **8858****SURVEY REPORT**

Customer: **Oasis Petroleum**
 Well Name: **Carson SWD 5301 12-24**
 Rig #: **Nabors B27**
 API #: **33-053-90329**
 Calculation Method: **Minimum Curvature Calculation**

MWD Operator: **D. Foley / D. Ogden**
 Directional Drillers: **RPM**
 Survey Corrected To: **True North**
 Vertical Section Direction: **180**
 Total Correction: **8.20**
 Temperature Forecasting Model (Chart Only): **Logarithmic**

Survey #	MD	Inc	Azm	Temp	TVD	VS	N/S	E/W	DLS
61	5764	72.10	181.30	122.00	5485.04	621.16	-621.16	-28.77	1.36
62	5858	71.50	179.90	129.00	5514.40	710.45	-710.45	-29.71	1.55
63	5952	72.10	179.70	131.00	5543.76	799.75	-799.75	-29.40	0.67
64	6047	71.40	178.50	132.00	5573.51	889.96	-889.96	-27.98	1.41
65	6141	70.30	178.50	134.00	5604.35	978.72	-978.72	-25.66	1.17
66	6235	70.00	178.40	134.00	5636.27	1067.11	-1067.11	-23.26	0.33
67	6329	69.90	178.50	134.00	5668.50	1155.38	-1155.38	-20.88	0.15
68	6424	71.30	178.20	136.00	5700.05	1244.94	-1244.94	-18.29	1.50
69	6518	71.20	178.20	134.00	5730.26	1333.91	-1333.91	-15.50	0.11
70	6611	72.30	178.40	138.00	5759.39	1422.19	-1422.19	-12.88	1.20
71	6705	72.80	178.40	140.00	5787.58	1511.83	-1511.83	-10.38	0.53
72	6799	69.80	178.40	141.00	5817.71	1600.82	-1600.82	-7.89	3.19
73	6894	71.10	178.40	143.00	5849.50	1690.31	-1690.31	-5.39	1.37
74	6988	71.80	178.00	143.00	5879.40	1779.38	-1779.38	-2.59	0.85
75	7082	72.80	177.70	143.00	5907.98	1868.87	-1868.87	0.77	1.11
76	7163	69.10	178.30	145.00	5934.42	1945.37	-1945.37	3.45	4.62
Projection	7217	69.10	178.30	145.00	5953.68	1995.80	-1995.80	4.94	0.00

**WELL COMPLETION OR RECOMPLETION REPORT - FORM 6**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 2468 (04-2010)



Well File No.
90329

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Designate Type of Completion				
☐ Oil Well	☐ EOR Well	☐ Recompletion	☐ Deepened Well	☐ Extended Horizontal Leg
☐ Gas Well	☒ SWD Well	☐ Water Supply Well	☐ Other:	
Well Name and Number Carson SWD 5301 12-24			Spacing Unit Description	
Operator Oasis Petroleum North America		Telephone Number (281) 404-9591	Field Baker	
Address 1001 Fannin, Suite 1500			Pool Dakota	
City Houston	State TX	Zip Code 77002	Permit Type ☐ Wildcat ☒ Development ☐ Extension	

LOCATION OF WELL

At Surface 584 F N L	2477 F WL	Qtr-Qtr NENW	Section 24	Township 153 N	Range 101 W	County McKenzie
Spud Date April 13, 2015	Date TD Reached April 19, 2015	Drilling Contractor and Rig Number Nabors B27			KB Elevation (Ft) 2109	Graded Elevation (Ft) 2084
Type of Electric and Other Logs Run (See Instructions) MWD/GR from surface to TD						

CASING & TUBULARS RECORD (Report all strings set in well)

Well Bore	String Type	Size (Inch)	Top Set (MD Ft)	Depth Set (MD Ft)	Hole Size (Inch)	Weight (Lbs/Ft)	Anchor Set (MD Ft)	Packer Set (MD Ft)	Sacks Cement	Top of Cement
Surface Hole	Surface	9 5/8	0	2210	13 1/2	36			812	0
Vertical Hole	Intermediate	7	0	7193	8 3/4	32			835	2180

PERFORATION & OPEN HOLE INTERVALS

Well Bore	Well Bore TD Drillers Depth (MD Ft)	Completion Type	Open Hole/Perforated Interval (MD,Ft)		Kick-off Point (MD Ft)	Top of Casing Window (MD Ft)	Date Perf'd or Drilled	Date Isolated	Isolation Method	Sacks Cement
Lateral1	7217	Perforations	5950	7080	4695		05/01/2015			

PRODUCTION

Current Producing Open Hole or Perforated Interval(s), This Completion, Top and Bottom, (MD Ft) Lateral 1-							Name of Zone (If Different from Pool Name)		
Date Well Completed (SEE INSTRUCTIONS) June 18, 2015		Producing Method		Pumping-Size & Type of Pump			Well Status (Producing or Shut-In)		
Date of Test	Hours Tested	Choke Size /64	Production for Test	Oil (Bbls)	Gas (MCF)	Water (Bbls)	Oil Gravity-API (Corr.)	Disposition of Gas	
Flowing Tubing Pressure (PSI)		Flowing Casing Pressure (PSI)		Calculated 24-Hour Rate	Oil (Bbls)	Gas (MCF)	Water (Bbls)	Gas-Oil Ratio	

GEOLOGICAL MARKERS

[illegible]

PLUG BACK INFORMATION

[illegible]

CORES CUT

Top (Ft)	Bottom (Ft)	Formation	Top (Ft)	Bottom (Ft)	Formation

Drill Stem Test

Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								

Well Specific Stimulations

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units Barrels
Type Treatment Sand Frac	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

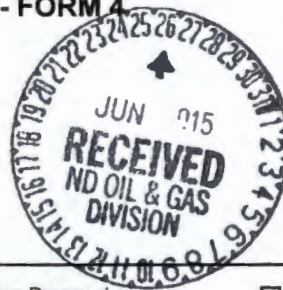
ADDITIONAL INFORMATION AND/OR LIST OF ATTACHMENTS

This is a preliminary completion report. A supplemental report will be filed upon first production of the well.

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.	Email Address	Date
	jswenson@oasispetroleum.com	06/25/2015
Signature	Printed Name	Title
	Jennifer Swenson	Regulatory Specialist

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.
90329

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☐ Notice of Intent	Approximate Start Date
☒ Report of Work Done	Date Work Completed June 18, 2015
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Injection Start Date for SWD Well

Well Name and Number CARSON SWD 5301 12-24					
Footages	Qtr-Qtr	Section	Township	Range	
584 F N L 2477 F W L	NENW	24	153 N	101 W	
Field Baker	Pool Dakota	County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum North America, LLC respectfully informs the NDIC that first injection for the subject Salt Water Disposal Well occurred on June 18, 2015.

** Submit a Form 6 Well Completion Report (KCC)*

Company Oasis Petroleum North America, LLC		Telephone Number (713) 770-6430	
Address 1001 Fannin Suite 1500			
City Houston	State TX	Zip Code 77002	
Signature <i>David Copeland</i>	Printed Name David Copeland		
Title Regulatory Specialist	Date June 24, 2015		
Email Address dcopeland@oasispetroleum.com			

FOR STATE USE ONLY	
☒ Received *	☐ Approved
Date 6/26/15	
By <i>Kevin Connors</i>	
Title KEVIN CONNORS UNDERGROUND INJECTION SUPERVISOR	

Well Integrity Report - Form 19 Summary

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840

Well File No. **90329**
UIC No. **W0437S0919D**

Test Date: **5/5/2015 11:13:32 AM** Contractor Performing Pressure Test: **Castle Oilfield Service**
Operator: **OASIS PETROLEUM NORTH AMERICA LLC** Telephone : **2814049500**
Address: **1001 FANNIN STE 1500** City: **HOUSTON** State: **TX** Zip Code: **77002**
Well Name: **CARSON SWD 5301 12-24** Field: **BAKER**
Location: Qtr-Qtr: **NENW** Section: **24** Township: **153** Range: **101** Cnty: **MCKENZIE**

WELL TEST DATA

Formation:	DAKOTA		Perforations:	5950 - 7080 Feet	
Tool Type:	PKR	Depth: 5821 Feet	Packer Model:	Nickle plated Arrowset	
Tubing Size:	4.5 Inches		Tubing Type:	11.6 # J-55 Polycore	
Well Type:	Disposal Well	Reason for Test:	UIC MIT	Test Type :	Initial MIT
BEFORE TEST START of TEST END of TEST					
Tubing Pressure:	0 PSIG	Shut In	0 PSIG	Shut In	0 PSIG
Annulus or Casing Pressure:	0 PSIG		1000 PSIG		960 PSIG
Annulus or Casing Fluid :	Treated Salt Water		Test Fluid:	Treated Salt Water	
Amount of Fluid Needed to Fill Annulus or Casing:	1.25 Bbls				
Test Length: 15 Minutes	Was Annulus or Casing Bled to Zero After Test?:		Yes	If No - Pressure Left: 	

INSPECTION COMMENTS

Test Result:	Acceptable	Report of Work Done Required:	Yes
MIT passed. Perfs, 7040-7010, 6970-7010, 6879-6894, 6820-6840, 6664-6679, 6597-6612, 6140-6170, 5950-5990. Pressure test three times.			
Submit Form 6 Completion Report. Due within 30 days after well completion (KCC)			
Submit Form 4 Sundry Notice date of first injection when applicable (KCC)			

This Report is true and complete to the best of my knowledge.

Company Representative Witnessing Test: **Josh Trowbridge** Title: **Rig Supervisor**
Commission Field Inspector Witnessing Test: **Richard Ryan**

NDIC Initials

KCC

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.

90329

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date April 15, 2015
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Change to Original APD

Well Name and Number Carson SWD 5301 12-24					
Footages		Qtr-Qtr	Section	Township	Range
584 F N L 2477 F W L		NENW	24	153 N	101 W
Field	Pool	County			
Baker	Dakota	McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum respectfully requests approval to make the following changes to the above permit:

Surface casing design:
Intermediate Casing of 7" with weight of 32# (previously weight 26#)

Company Oasis Petroleum North America LLC		Telephone Number (281) 404-9500	
Address 1001 Fannin Suite 1500			
City Houston		State TX	Zip Code 77002
Signature <i>VS</i>	Printed Name Victoria Siemieniewski		
Title Regulatory Specialist	Date March 23, 2015		
Email Address vsiemieniewski@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 3/30/2015	
By <i>Kevin Connors</i>	
Title KEVIN CONNORS	
UNDERGROUND INJECTION SUPERVISOR	



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.dmr.nd.gov/oilgas/

PERMIT FOR FLUID INJECTION

OPERATOR: Oasis Petroleum North America, LLC

ADDRESS: 1001 Fannin, Suite 1500 Houston, TX 77002

WELL NAME & NO.: Carson SWD 5301 12-24

LOCATION: NENW Section 24, T153N, R101W McKenzie County, ND

AUTHORIZED INJECTION ZONE: Dakota Group

UNDERGROUND INJECTION CONTROL (UIC) NO.: W0437S0919D

WELL FILE NO.: 90329

PERMIT DATE: March 20, 2015

The operator shall notify the North Dakota Industrial Commission ("Commission") at least 24 hours prior to plugging back and converting said well to a saltwater disposal well. This permit is conditioned upon the operator complying with the provisions set forth in Chapter 43-02-05, Chapter 43-02-03, any other applicable rules or orders of the Commission, and the following stipulations:

(1) Effective June 1, 2014, all operators of disposal wells in North Dakota will be required to maintain an on-site container to store filters until they can be properly disposed of in an authorized facility. Such containers must be:

- Leak-proof to prevent any fluids from escaping
- Covered to prevent precipitation from entering the container
- Labeled to indicate only filters are to be placed in the container

(2) The operator shall obtain approval on plug-back procedure prior to converting said well to a saltwater disposal well.

(3) The injection interval shall be isolated by cement and verified by a cement bond log prior to injecting fluids.

(4) Injection shall be through tubing and packer set within 100 feet of the top perforation.

(5) A Commission Field Inspector must witness a satisfactory mechanical integrity test on the tubing-casing annulus prior to injecting fluids.

(6) Accurate pressure gauges shall be placed on the tubing and on the tubing-casing annulus.

(7) Written notice must be given of the date of first injection immediately and of any discontinuance within ten days. Written notice must also be given of the date injection recommences.

(8) All fluids injected shall be metered or measured using an approved method.

(9) The operator is only authorized to inject Class II fluids into said well. Class II fluids are generally defined as follows:

- a. Produced water from oil and gas production;
- b. Waste fluids from actual drilling operations;
- c. Used workover, completion, and stimulation fluids recovered from production, injection and exploratory wells;
- d. Waste fluids from circulation during well cementing;
- e. Pigging fluids from cleaning of collection and injection lines within the field;
- f. Enhanced recovery waters including fresh water makeup and other waters containing chemicals for the purpose of enhanced recovery;
- g. Gases used for enhanced recovery/pressure maintenance of production reservoirs;
- h. Brine reject from water softeners associated with enhanced recovery;
- i. Waste oil and fluids from clean up within the oil field;
- j. Waste water from gas plants which are an integral part of production operations unless those waters are classified as hazardous waste at the time of injection.

(10) Copies of all logs run hereinafter and a Sundry Notice (Form 4) detailing the conversion of this well must be submitted to the Commission within 30 days after the completion of the conversion. All logs run must be submitted free of charge, as one digital TIFF (tagged image file format) copy and one digital LAS (log ASCII) formatted copy. The Sundry Notice shall include, but not be limited to, the following information: work completed and depths to plug back well, plug back depth, details of any isolation squeezes, perforated interval, packer type and depth, tubing size and type, and any other remedial work done as well as the dates the work was performed

(11) The operator shall report any noncompliance with regulations or permit conditions within 24 hours followed by a written explanation within five days.

(12) Approval shall be obtained prior to attempting any remedial work to the casing strings.

(13) Following any remedial work done to this well over the life of this well, a Sundry Notice (Form 4) must be submitted detailing the work done. This report shall include the reason for the workover, the work done, dates, size and type of tubing, type and location of packer, result of pressure test, and any other pertinent data.

(14) Following the completion of any remedial work, the Director is authorized to order further mechanical integrity tests or other remedial work as deemed necessary to insure the mechanical integrity of the well and to prevent the movement of injected fluids into an underground source of drinking water or a zone other than authorized herein.

(15) The operator shall report monthly, regardless of the status of the operations, the amount of any fluid injected, the sources thereof, and the average injection pressure. The reports (Forms 16 and 16A) shall be filed with the Director on or before the fifth day of the second month following that in which injection occurred or could have occurred.

(16) If oil is recovered from the disposal operations, the operator shall report monthly, regardless of the status of the operations, the amount of oil recovered and sold, and the purchaser of said oil. The reports (Form 5 SWD) shall be filed with the Director on or before the 1st day of the second succeeding month.

- (17) The well and injection system shall be kept under continuing surveillance.
- (18) Operations shall cease immediately if so directed by the Director or one of his representatives.
- (19) The operator shall conduct tests and install monitoring equipment as prescribed by the Director to verify the integrity of the surface facility, gathering system and injection well and to insure that the surface and subsurface potable waters will be protected.
- (20) All fires, breaks, leaks, spills, or other accidents of this nature shall be reported to the Oil and Gas Division pursuant to North Dakota Administrative Code (NDAC) Section 43-02-03-30.
- (21) The Director is hereby authorized to issue administrative orders to include such other terms and conditions as may be necessary to insure proper compliance with the provisions of North Dakota Century Code Chapter 38-08 of the and NDAC Chapter 43-02-05 and 43-02-03. The Director is further authorized to immediately order the well shut-in if mechanical failure or downhole problems indicate that injected fluids are, or may be, moving into a zone other than authorized herein, or if the operator of the well fails to comply with the statutes, rules, orders of the Commission or written or oral directives of the Commission or its staff.
- (22) Unless the well is converted to a saltwater disposal well by March 20, 2016, this permit shall expire and be of no further force and effect.
- (23) If the authorized injection zone is plugged and abandoned, this permit shall expire and be of no further force and effect.

This permit is effective the date set forth below and is transferable only upon approval of the Commission.

Dated this 20th day of March, 2015.

North Dakota Industrial Commission
Oil and Gas Division
Lynn D. Helms
Director

By: 
Kevin C. Connors
UIC Supervisor

UIC PERMIT CHECKLIST

Well File No.
90329

Operator Oasis Petroleum				Well Name Carson SWD 5301 12-24			Date Received December 12, 2014	
Location of Well 584 F N L 2477 F W L NENW				Section 24	Township 153 N	Range 101 W	County McKenzie	
Field Baker				Current Pool Dakota			Current Status Proposed	
Logs on File none				Top of Cement 2159 Feet		TOC Determined By and Date ☒ CBL ☐ Temp ☐ Calc		
Injection Zone Dakota (Inyan Kara)		Top 5421 Feet		Confining Zone Mowry			Thickness 153 Feet	

- ☒ 1. Operator and well on bond as listed.
- ☒ 2. Completed Form 14.
Estimated frac pressure of confining zone. 4337 psi, gradient 0.8 psi/ft.
Maximum injection rate and pressure. 15000 bpd @ 1450 psi.
K_{FB}base 1987 Feet.
- ☒ 3. Lithology of proposed injection zone and confining zones.*
- ☒ 4. Map of area of review.
- ☒ 5. Name and location of wells in area of review.
- ☒ 6. Corrective action on wells in area of review.
- ☒ 7. Injection program.*
- ☒ 8. ☒ a. Analysis from state certified lab of two nearest fresh water wells.*
_____ b. For units listed below: name and location of two nearest fresh water wells.
_____ c. For units listed below: analysis from state certified lab of any new fresh water wells.
- ☒ 9. Analysis from state certified lab of injection fluid.*
- ☒ 10. List of source wells.*
- ☒ 11. Landowner description within area of review.*
- ☒ 12. Certification of notification of landowners and letter sent.*
- ☐ 13. Logs and test data if not previously submitted.
- ☒ 14. Drawing of current* and proposed well bore and surface construction.*
- ☒ 15. Dike Schematic
- ☒ 16. Traffic Flow Diagram
- ☒ 17. Sundry detailing conversion procedure.
- ☒ 18. Form 1
- ☒ 19. Shallow aquifer check.
- ☒ 20. Legal Street Address Requested

Date Landowner Notified
December 2, 2014

Note: "*" indicates that the item is not required for EOR wells in the Cedar Creek, Cedar Hills North, Medicine Pole Hills South, Mohall, and Wiley units.

Hearing

Hearing Date January 21, 2015	Case No. 23553	Order No. 25893	Permit Date March 20, 2015	UIC Number W0437S0919D
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Administrative

Order No. Allowing Injection na	Permit Date na	UIC Number na
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Permit Approved By
Kevin Connors (Initials)
KCC

AREA OF REVIEW DATA SHEET

(WELLS WITHIN 1/4 MILE RADIUS PENETRATING INJECTION ZONE)
(05-2000)

Well File No.
90329

Well Name and Number Carson SWD 5301 12-24						Number of Wells in AOR 4	
Footages 584 F N L 2477 F W L		Qtr-Qtr NENW	Section 24	Township 153 N	Range 101 W	County McKenzie	
Field Baker				Injection Zone Dakota (Inyan Kara)		Injection Zone Top 5421 Feet	

AREA OF REVIEW WELLS

Well Name and Number Yukon #5301 41-12T				Status A	Status Date 9/9/2013	Well File No. 22099	
Footages 255 F S L 710 F W L		Qtr-Qtr SWSW	Section 12	Township 153 N	Range 101 W	County McKenzie	
Surface Casing 9 5/8 "	Surface Casing Depth 2127 Feet	Cement Volume 747 Sacks	Top of Cement 0	Top of Cement Determined By Visual			
Production Casing 7 "	Production Casing Depth 11126 Feet	Cement Volume 794 Sacks	Top of Cement 2563 Feet	Top of Cement Determined By ☒ CBL ☐ Temp ☐ Calc			
Injection Zone Three Forks		Injection Zone Top 10794 Feet	Pierre Shale Top 1986 Feet	Logs on File CBL, CIL, DTSM			
Comments Three Forks Lateral leg in AOR - Geological isolation from injection zone							

Well Name and Number Jefferies #5301 43-12B				Status A	Status Date 3/22/2012	Well File No. 22220	
Footages 250 F S L 2510 F E L		Qtr-Qtr SWSW	Section 12	Township 153 N	Range 101 W	County McKenzie	
Surface Casing 9 5/8 "	Surface Casing Depth 2170 Feet	Cement Volume 630 Sacks	Top of Cement 0	Top of Cement Determined By Visual			
Production Casing 7 "	Production Casing Depth 11093 Feet	Cement Volume 755 Sacks	Top of Cement 3236 Feet	Top of Cement Determined By ☒ CBL ☐ Temp ☐ Calc			
Injection Zone Bakken		Injection Zone Top 10740 Feet	Pierre Shale Top 2018 Feet	Logs on File CBL, DTSM			
Comments Middle Bakken Lateral leg in AOR - Geological isolation from injection zone							

Well Name and Number Innoko #5301 43-12T				Status A	Status Date 9/18/2013	Well File No. 22221	
Footages 250 F S L 2350 F E L		Qtr-Qtr SWSE	Section 12	Township 153 N	Range 101 W	County McKenzie	
Surface Casing 9 5/8 "	Surface Casing Depth 2152 Feet	Cement Volume 691 Sacks	Top of Cement 0	Top of Cement Determined By Visual			
Production Casing 7 "	Production Casing Depth 11117 Feet	Cement Volume 793 Sacks	Top of Cement 3330 Feet	Top of Cement Determined By ☒ CBL ☐ Temp ☐ Calc			
Injection Zone Three Forks		Injection Zone Top 10811 Feet	Pierre Shale Top 2002 Feet	Logs on File CBL, CIL, DTSM			
Comments Three Forks Lateral leg in AOR - Geological isolation from injection zone							

AREA OF REVIEW DATA SHEET

(WELLS WITHIN 1/4 MILE RADIUS PENETRATING INJECTION ZONE)
(05-2000)

Well File No.
90329

Well Name and Number Carson SWD 5301 12-24							Number of Wells in AOR 4	
Footages 584 F N L 2477 F W L		Qtr-Qtr NENW	Section 24	Township 153 N	Range 101 W	County McKenzie		
Field Baker				Injection Zone Dakota (Inyan Kara)		Injection Zone Top 5421 Feet		

AREA OF REVIEW WELLS

Well Name and Number Bray #5301 43-12H					Status A	Status Date 1/13/2012	Well File No. 20864
Footages 250 F S L 1927 F E L		Qtr-Qtr SWSE	Section 12	Township 153 N	Range 101 W	County McKenzie	
Surface Casing 9 5/8 "	Surface Casing Depth 2089 Feet	Cement Volume 605 Sacks	Top of Cement 0	Top of Cement Determined By Visual			
Production Casing 7 "	Production Casing Depth 11071 Feet	Cement Volume 785 Sacks	Top of Cement 3165 Feet	Top of Cement Determined By ☒ CBL ☐ Temp ☐ Calc			
Injection Zone Bakken		Injection Zone Top 10719 Feet	Pierre Shale Top 1990 Feet	Logs on File CBL, DTSM, HDL, HRL, HSL			
Comments Middle Bakken Lateral leg in AOR - Geological isolation from injection zone							

Well Name and Number					Status	Status Date	Well File No.
Footages F L F L		Qtr-Qtr	Section	Township N	Range W	County	
Surface Casing "	Surface Casing Depth Feet	Cement Volume Sacks	Top of Cement	Top of Cement Determined By			
Production Casing "	Production Casing Depth Feet	Cement Volume Sacks	Top of Cement Feet	Top of Cement Determined By ☒ CBL ☐ Temp ☐ Calc			
Injection Zone		Injection Zone Top Feet	Pierre Shale Top Feet	Logs on File			
Comments							

Well Name and Number					Status	Status Date	Well File No.
Footages F L F L		Qtr-Qtr	Section	Township N	Range W	County	
Surface Casing "	Surface Casing Depth Feet	Cement Volume Sacks	Top of Cement	Top of Cement Determined By			
Production Casing "	Production Casing Depth Feet	Cement Volume Sacks	Top of Cement Feet	Top of Cement Determined By ☒ CBL ☐ Temp ☐ Calc			
Injection Zone		Injection Zone Top Feet	Pierre Shale Top Feet	Logs on File			
Comments							

**APPLICATION FOR INJECTION - FORM 14**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 18669 (08-2012)



PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM. PLEASE SUBMIT THE ORIGINAL AND TWO COPIES.
APPROVAL MUST BE OBTAINED BEFORE WORK COMMENCES.

Permit Type ☐ Enhanced Recovery ☒ Saltwater Disposal ☐ Gas Storage		Injection Well Type ☐ Converted ☒ Newly Drilled		Commercial SWD ☐ Yes ☒ No	
Operator Oasis Petroleum North America LLC		Telephone Number (281) 404-9500		Will Oil be Skimmed? ☒ Yes ☐ No	
Address 1001 Fannin St, Suite 1500		City Houston		State TX	Zip Code 77007
Well Name and Number Carson SWD 5301 12-24		Field or Unit Baker			

LOCATIONS

At Surface 584 F N L 2477 F W L	Qtr-Qtr NENW	Section 24	Township 153 N	Range 101 W	County McKenzie
Bottom Hole Location 2584 F N L 2477 F W L	Qtr-Qtr NENW	Section 24	Township 153 N	Range 101 W	County McKenzie

Geologic Name of Injection Zone Dakota Group		Top 5421 Feet	Injection Interval 5736-7196 MD gross interval (5489-5863 TVD) Feet	
Geologic Name of Top Confining Zone Skull Creek		Thickness 250 Feet	Geologic Name of Bottom Confining Zone Swift Thickness 350 Feet	
Bottom Hole Fracture Pressure of the Top Confining Zone 4337 PSI		Gradient 0.80 PSI/ft		
Estimated Average Injection Rate and Pressure 10000 BPD @ 1000 PSI		Estimated Maximum Injection Rate and Pressure 15000 BPD @ 1450 PSI		
Geologic Name of Lowest Known Fresh Water Zone Fox Hill			Depth to Base of Fresh Water Zone 2059 Feet	
Total Depth of Well (MD & TVD) 7254 MD, 5878 TVD Feet		Logs Previously Run on Well triple combo through mowry, GR/Res to BSC, GR to surf		

CASING, TUBING, AND PACKER DATA (Check If Existing)

NAME OF STRING	SIZE	WEIGHT (Lbs/Ft)	SETTING DEPTH	SACKS OF CEMENT	TOP OF CEMENT	TOP DETERMINED BY
Surface ☒	9 5/8"	36	2159	635	Surface	Visual
Intermediate ☒	7"	26	7254	509	BSC	CBL
Long String ☐						

Liner ☐	TOP	BOTTOM	SACKS OF CEMENT

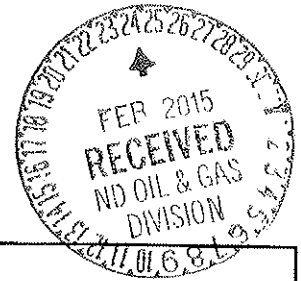
Proposed Tubing	SIZE	WEIGHT	SETTING DEPTH	TYPE
	4 1/2"	10.5	5634	J-55 Sealite Coated

Proposed Packer Setting Depth 5604 Feet	Model Internally coated Baker lockset	☒ Compression ☐ Permanent ☐ Tension
---	---	--

FOR STATE USE ONLY

Permit Number and Well File Number 90329	
UIC Number W043750919D	Approval Date 3/20/2015
By 	
Title KEVIN CONNORS	

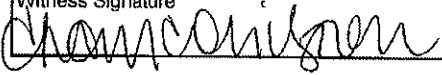
**UNDERGROUND INJECTION
SUPERVISOR**



COMMENTS

--

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.		Date February 24, 2015
Signature 	Printed Name Victoria Siemieniewski	Title Regulatory Specialist

Above Signature Witnessed By		
Witness Signature 	Witness Printed Name Chelsea Covington	Witness Title Regulatory Assistant

Instructions

1. Attach a list identifying all attachments.
2. The operator, well name and number, field or unit, well location, and any other pertinent data shall coincide with the official records on file with the Commission. If it does not, an explanation shall be given.
3. If an injection well is to be drilled, an Application for Permit to Drill - Form 1 (SFN 4615) shall also be completed and accompanied by a plat prepared by a registered surveyor and a drilling fee.
4. Attach a lithologic description of the proposed injection zone and the top and bottom confining zones.
5. Attach a plat depicting the area of review (1/4-mile radius) and detailing the location, well name, and operator of all wells in the area of review. Include: injection wells, producing wells, plugged wells, abandoned wells, drilling wells, dry holes, and water wells. The plat shall also depict faults, if known or suspected.
6. Attach a description of the needed corrective action on wells penetrating the injection zone in the area of review.
7. Attach a brief description of the proposed injection program.
8. Attach a quantitative analysis from a state-certified laboratory of fresh water from the two nearest fresh water wells. Include legal descriptions.
9. Attach a quantitative analysis from a state-certified laboratory of a representative sample of water to be injected.
10. Attach a list identifying all source wells, including location.
11. Attach a legal description of land ownership within the area of review. List ownership by tract or submit in plat form.
12. Attach an affidavit of mailing certifying that all landowners within the area of review have been notified of the proposed injection well. This notice shall inform the landowners that comments or objections may be submitted to the Commission within 30 days, or that a hearing will be held at which comments or objections may be submitted, whichever is applicable. Include copies of letters sent.
13. Attach all available logging and test data on the well which has not been previously submitted.
14. Attach schematic drawings of the injection system including current well bore construction and proposed well bore and surface facility construction.
15. Attach a Sundry Notice - Form 4 (SFN 5749) detailing the proposed procedure.
16. Attach a diagram representing the traffic flow and the maximum number of trucks staged on site.
17. Attach a printout of a map obtained at <http://www.nd.gov/gis/apps/HubExplorer/> with surficial aquifers (under hydrography) active, and proposed location plotted on printout.
18. Read Section 43-02-05-04 of the North Dakota Administrative Code to ensure that this application is complete.
19. The original and two copies of this application and attachments shall be filed with the Industrial Commission of North Dakota, Oil and Gas Division, 600 East Boulevard, Dept. 405, Bismarck, ND 58505-0840.



12/8/2014

Oasis Petroleum
1001 Fannin St.
Suite 1500
Houston, TX 77002



Kevin Connors
UIC/CCS Supervisor
North Dakota Industrial Commission
600 East Boulevard Avenue Dept. 405
Bismarck, ND 58505-0840

RE: Carson SWD 5301 12-24

Dear Mr. Connors:

Pursuant to NDCC 43-02-03-16, Oasis respectfully submits the **Carson 5301 SWD 12-24** well APD for your consideration. Included with the application are all necessary attachments; an itemized list is included on the following page.

This well is located in Section 24 of Township 153 N, Range 101 W. This location is part of Indian Hills, Baker field and is in a generally rural area. A regional map has been provided with this application. Additionally, there are no federal minerals associated with this well. The landowners, Wesley and Barbara Lindvig, have been fully apprised of Oasis' development plans. The drill cuttings and solids will be disposed of in an onsite pit. Liquids from the drilling process will be taken to an authorized disposal facility.

Please do not hesitate to contact me should you have any additional questions or concerns regarding this well. Thank you for your consideration.

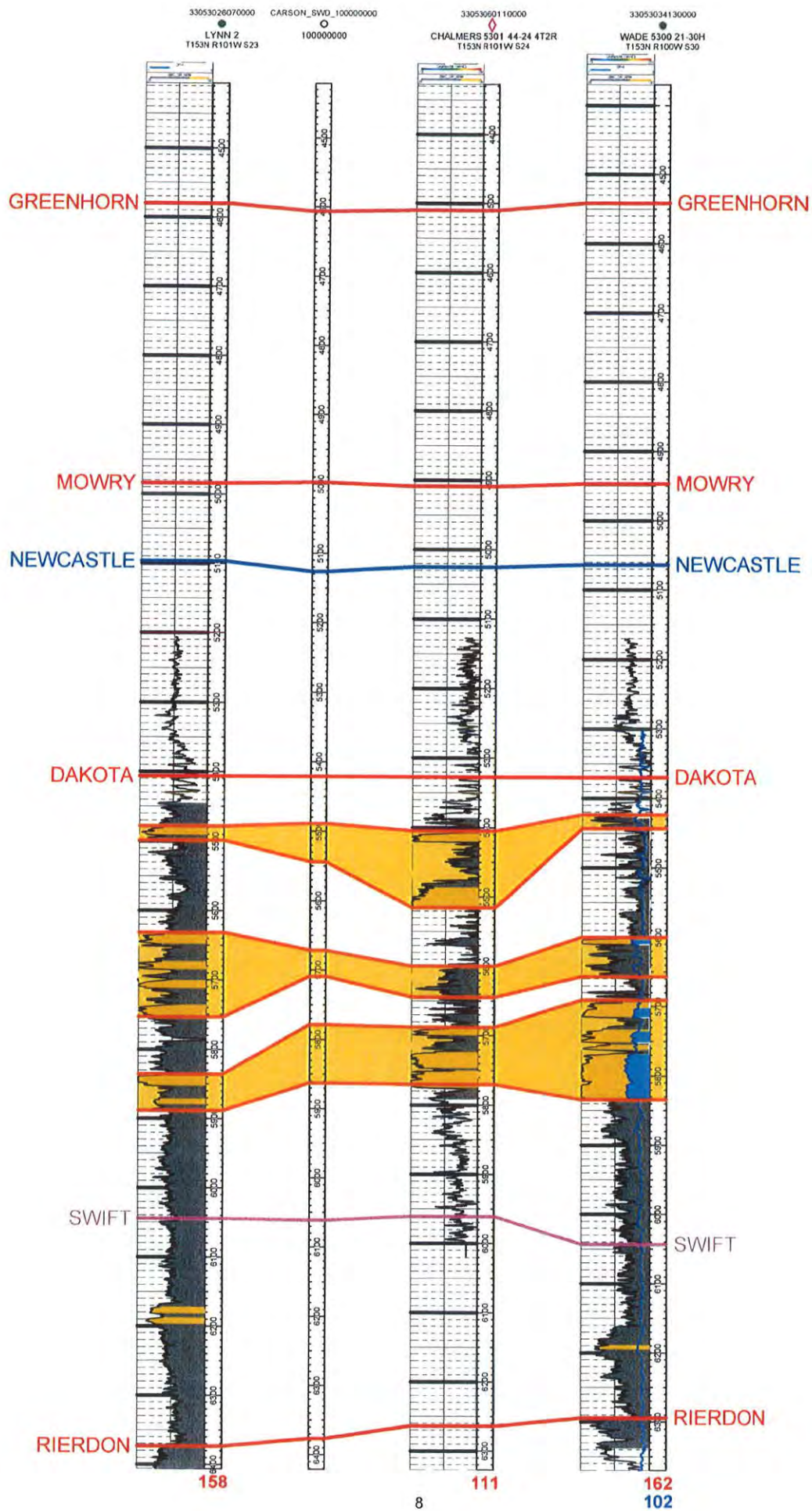
Respectfully,

A handwritten signature in black ink that reads "Michael Kukuk".

Michael Kukuk
Regulatory Supervisor
281.404.9575
Oasis Petroleum

Lithologic Description

- Well:** Carson SWD 5301 12-24
Salt Water Disposal Well
McKenzie County, North Dakota
24-153N-101W
- Injection Zone:** The Cretaceous Inyan Kara within the Dakota is roughly 440' in this area based on the Oasis Petroleum Wade 5300 21-30H. The section consists of a sequence of alternating sand, silt and shale. Several potential injection zones occur between 5,501' and 5,943' based on the Oasis Petroleum Wade 5300 21-30H. These zones vary in thickness between 10' to 80'.
- Upper Unit:** Shales of the Skull Creek and Newcastle units confine the upper boundary of the Inyan Kara Formation. Individual shale units over 60' thick are inter-bedded with thinner siltstones.
- Lower Unit:** The Swift Shale consists of fine siltstones in the upper portion that grade into shales in the lower half of the unit. This formation spans 350' based on the Oasis Petroleum Wade 5300 21-30H.



Proposed Injection Program

November 25, 2014

Carson SWD 5301 12-24
Oasis Petroleum North America, LLC
Section 24, T153N R101W
McKenzie County, North Dakota

Application for Fluid Injection

PROPOSED INJECTION PROGRAM

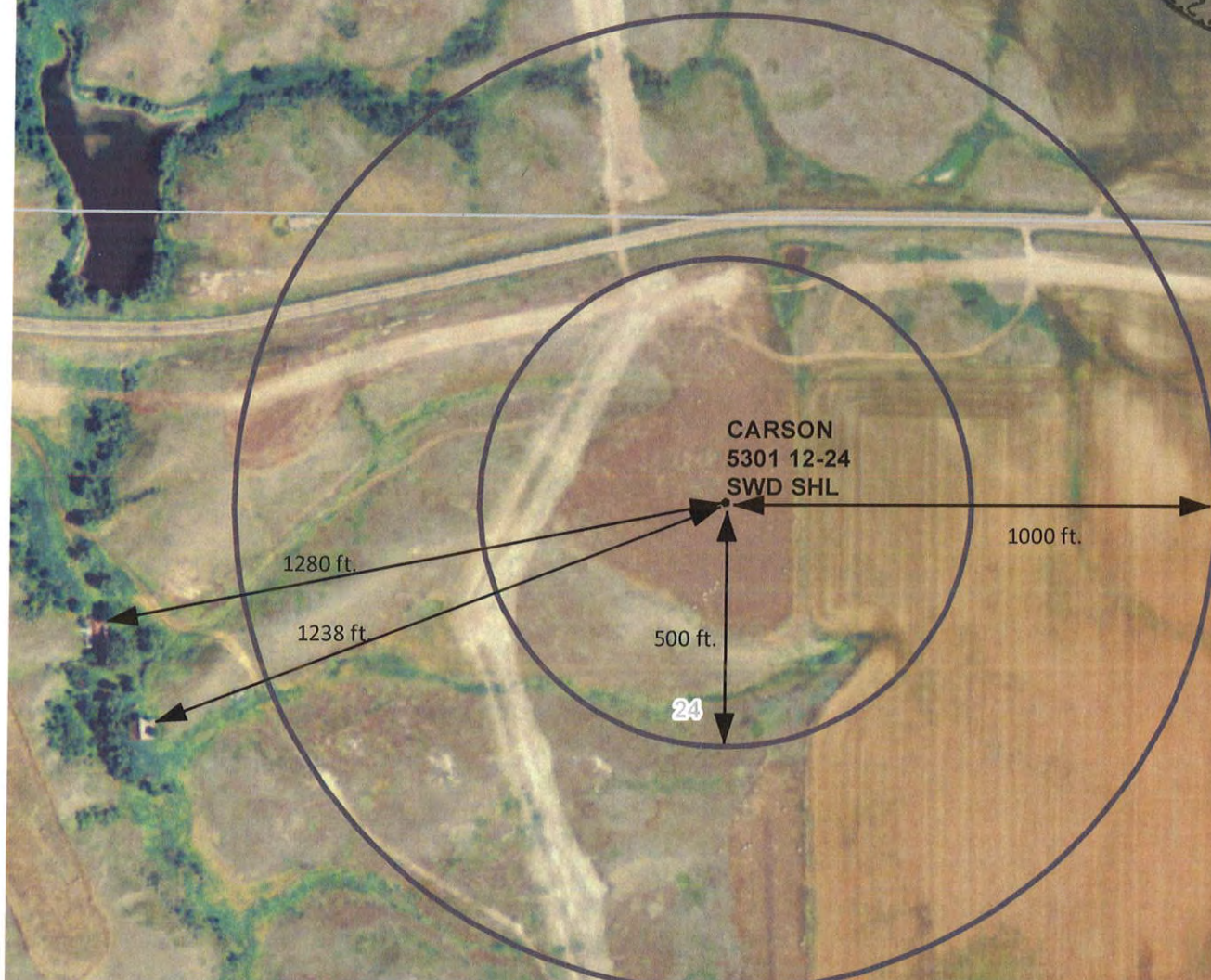
The Carson SWD 5301 12-24 well will be used to inject produced water from wells in the Baker Field-Oasis Bakken and Three Forks wells- already on production, in the process of completion, and those soon to be drilled in the four township area of 152-153N and 100-101W into the Dakota formation. Injection depths will be determined after the well is drilled and logged.

The source wells for the produced water are listed on Exhibit 6 but will also include additional Bakken and Three Forks wells yet to be drilled and completed. Water from these wells will be shipped via pipeline to the Carson SWD 5301 12-24 disposal site; trucking will only be done for emergency and/or unplanned circumstances. The amount of water injected into the Dakota is pressure dependent, but not to exceed ~~1,550~~ psig at surface.

1450

The proposed Carson SWD 5301 12-24 is needed to provide a more local disposal facility in the area to reduce costs and ease the burden on regional disposal wells. The anticipated initial injection rate of the well will be 10,000 bwpd at 1,000 psig pressure. However, this rate may increase with additional drilling in the area, subject to the maximum pressure ceiling, and maximum daily injection rate. The maximum daily injection rate is based on diking dimensions around tanks. Pursuant to NDAC section 43-02-03-53 (3)(b).

Exhibit B - Area of Review Plat



CARSON
5301 12-24
SWD SHL

1000 ft.

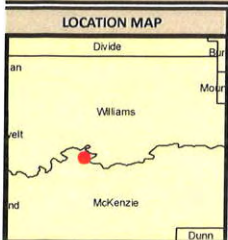
1280 ft.

1238 ft.

500 ft.

24

13

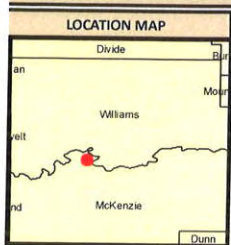


CARSON SWD 5301 43-21 12-24
153N - 101W, Section 24
McKenzie County, ND

- Proposed SWD Well
- Well Buffers



Exhibit B - Area of Review Plat



CARSON SWD 5301 43-21 12-24
153N - 101W, Section 24
McKenzie County, ND

• Proposed SWD Well
Well Buffers

OASIS
PETROLEUM

Connors, Kevin C.

From: Michael Kukuk <mkukuk@oasispetroleum.com>
Sent: Tuesday, December 16, 2014 12:52 PM
To: Connors, Kevin C.
Subject: FW: carson
Attachments: ESB_AerialOverview_CarsonSWD5301.pdf; ATT00001.txt

Hi Kevin –

Please see the attached aerial overview. Please let me know if you have any questions/concerns.

Thanks,
Michael

From: Victoria Siemieniewski
Sent: Tuesday, December 16, 2014 12:06 PM
To: Michael Kukuk
Subject: FW: carson

Hello Michael,

Please see attached Aerial overview and below GIS snapshot. The nearest homes are over 1200 ft away from the well head.

Sincerely,

Victoria

From: Eric Branton
Sent: Tuesday, December 16, 2014 11:20 AM
To: Victoria Siemieniewski
Subject: carson

Eric Branton

GIS Lead

Oasis Petroleum

1001 Fannin, Suite 1500

Houston, Texas 77002

281-404-9448

ebranton@oasispetroleum.com





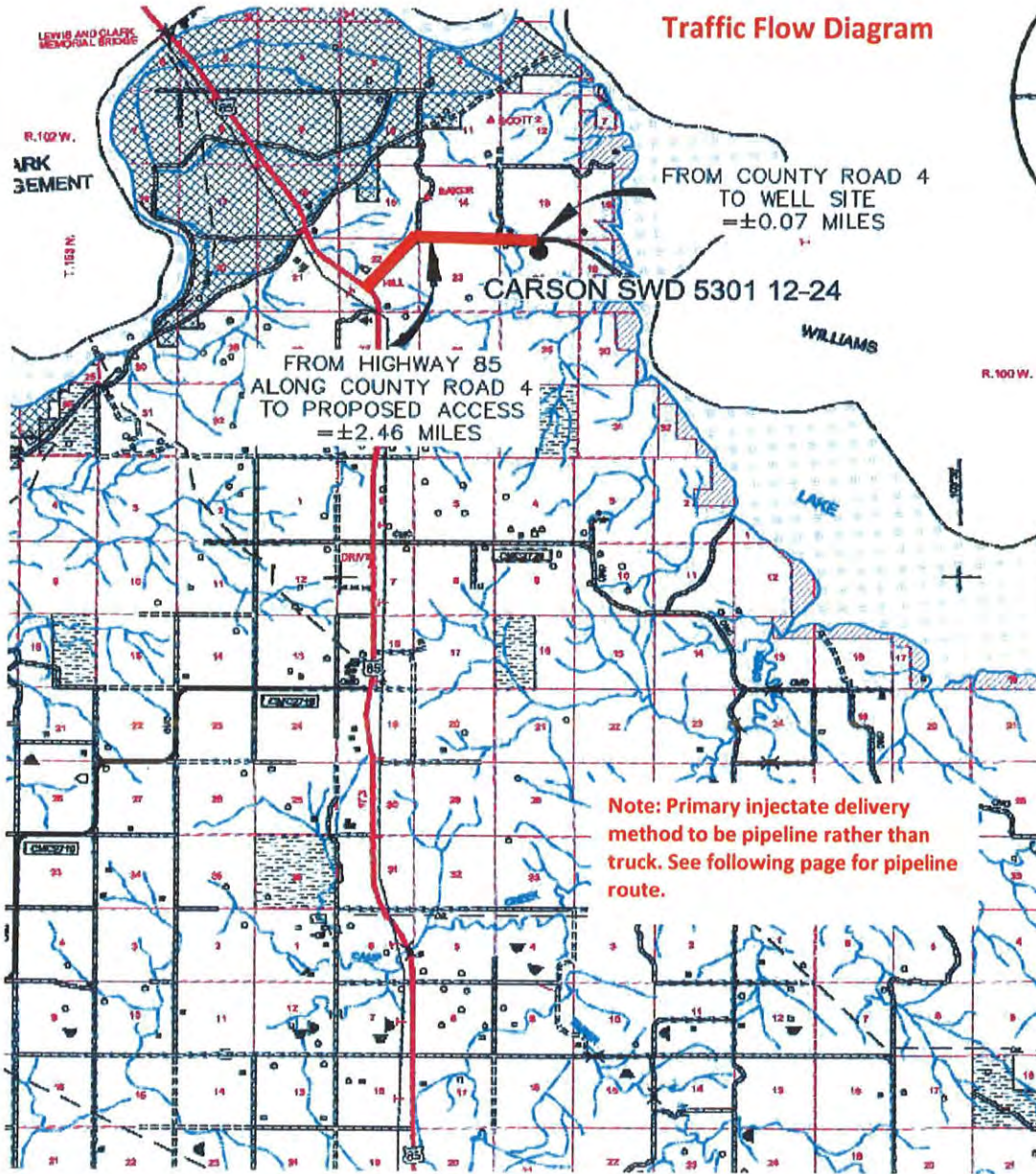
Here's approximate distances to all the structures...you just want the one to the northwest or the northeast too?



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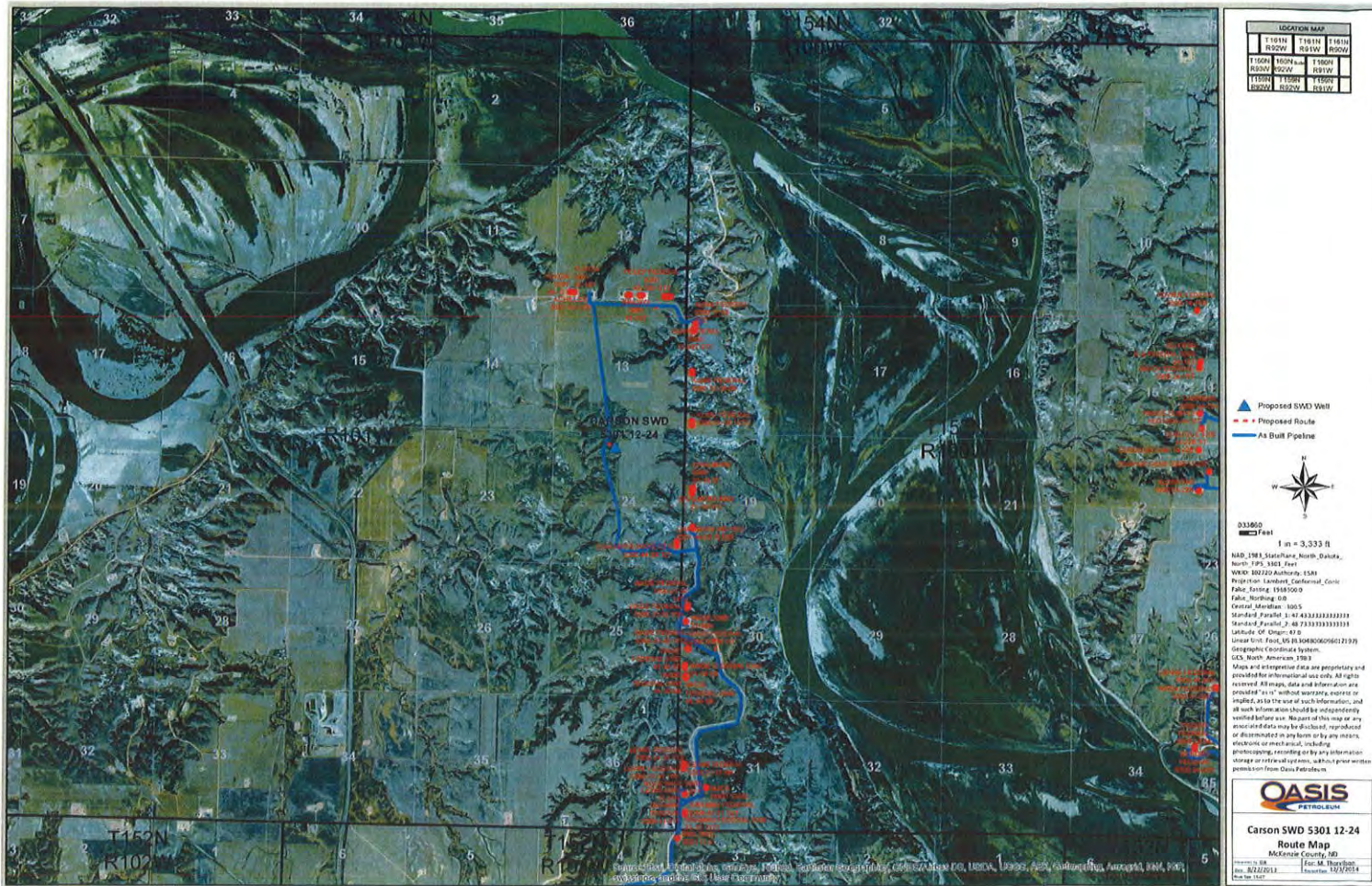
COUNTY ROAD MAP
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "CARSON SWD 5301 12-24"
 584 FEET FROM NORTH LINE AND 2477 FEET FROM WEST LINE
 SECTION 24, T153N, R101W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA

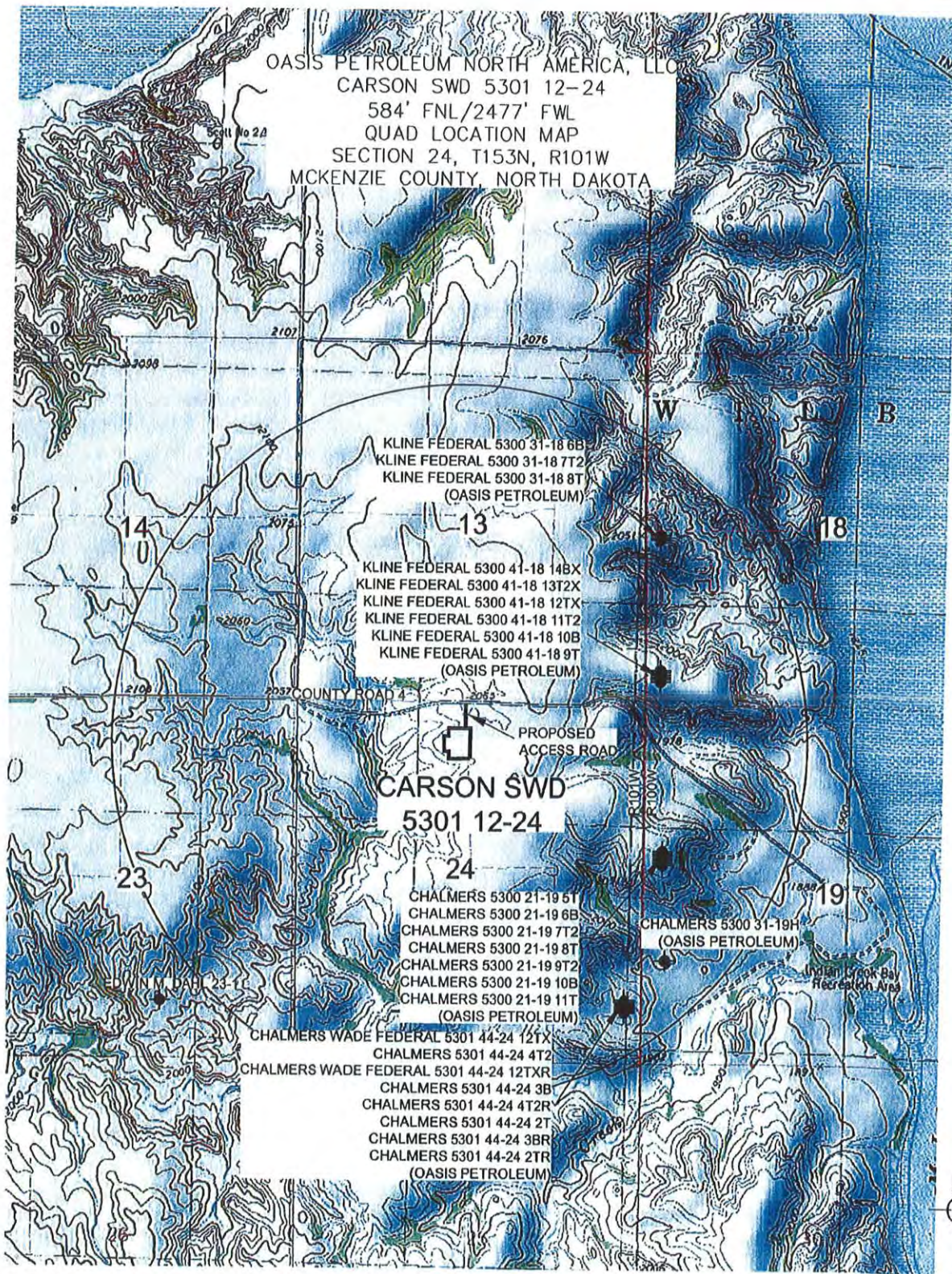


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SCALE: 1" = 2 MILE

INTERSTATE ENGINEERING Professional services you need, exactly your best.	Interstate Engineering, Inc. P.O. Box 640 425 East Main Street Bismarck, Montana 58103 Ph (406) 433-0617 Fax (406) 433-0610 www.interstate92.com	OASIS PETROLEUM NORTH AMERICA, LLC COUNTY ROAD MAP SECTION 24, T153N, R101W MCKENZIE COUNTY, NORTH DAKOTA		Drawn By: <u>BJP</u> Project No.: <u>51426-004</u> Checked By: <u>BJP</u> Date: <u>MAY 25, 14</u>												
		<table border="1"> <thead> <tr> <th>Rev</th> <th>Date</th> <th>Description</th> </tr> </thead> <tbody> <tr><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td></tr> </tbody> </table>		Rev	Date	Description										
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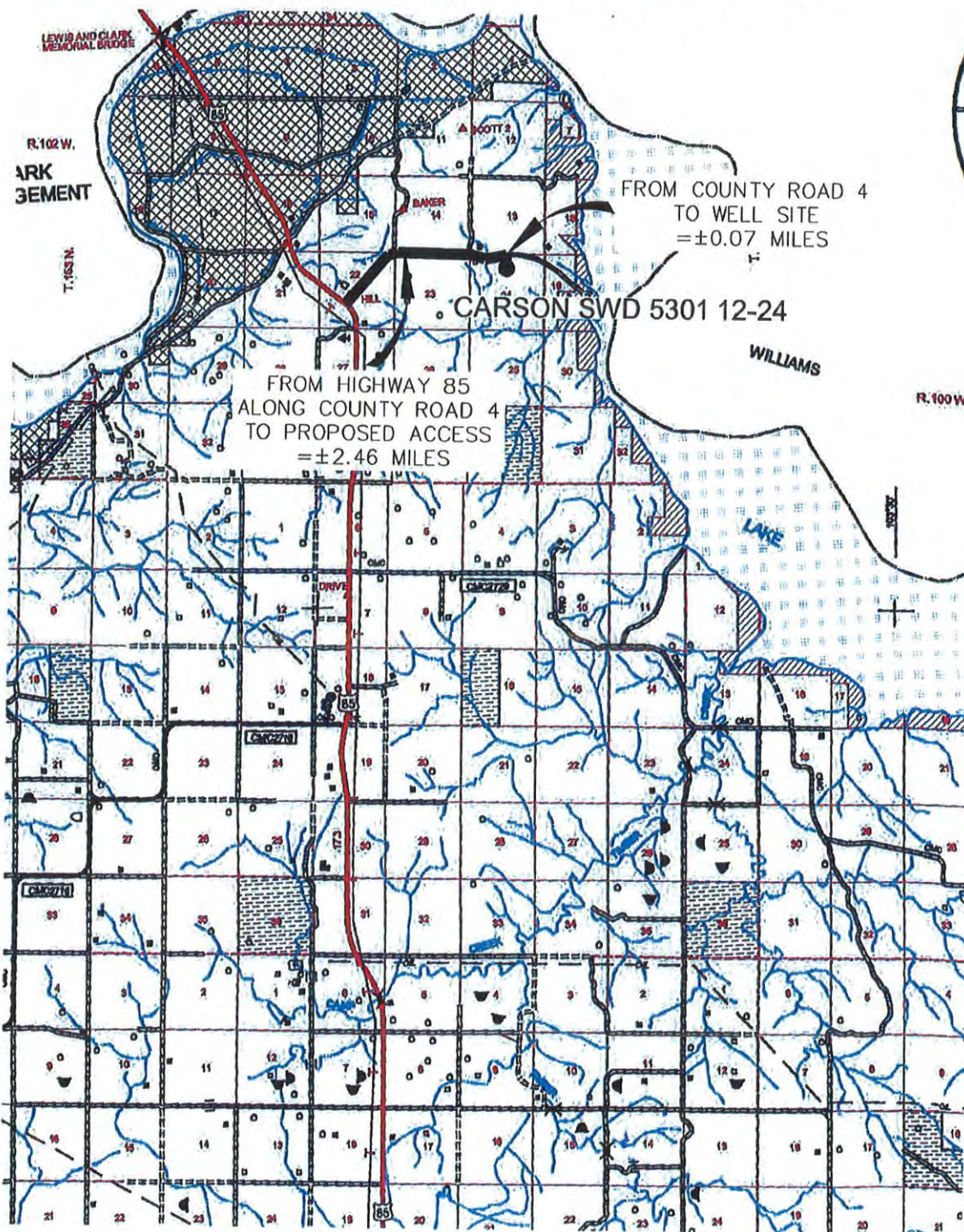
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425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.interstateeng.com
Our offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
QUAD LOCATION MAP
SECTION 24, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S14-09-254
Checked By: D.D.K. Date: NOV. 2014

Revised No.	Date	By	Description

COUNTY ROAD MAP
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "CARSON SWD 5301 12-24"
 584 FEET FROM NORTH LINE AND 2477 FEET FROM WEST LINE
 SECTION 24, T153N, R101W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA



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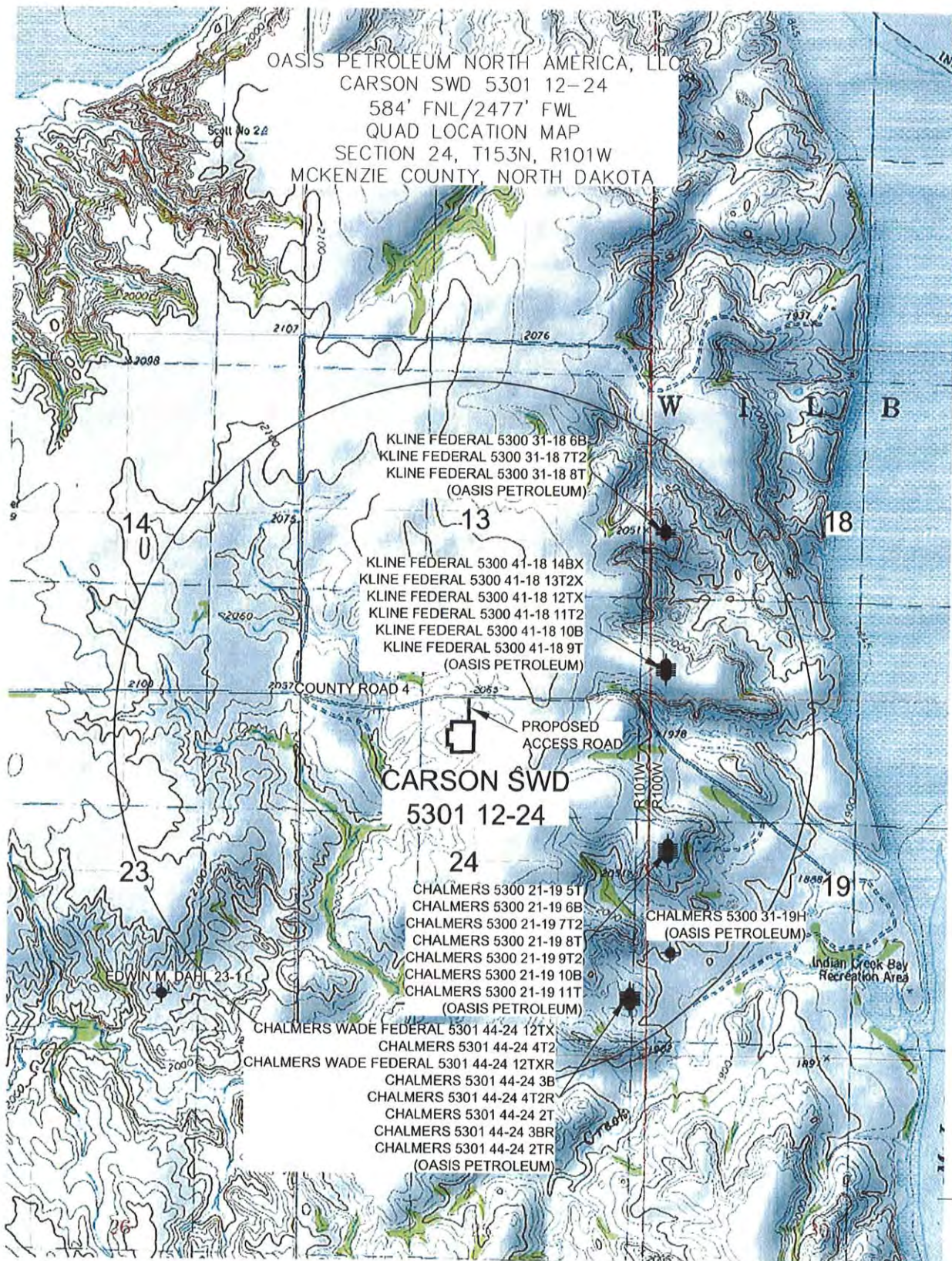


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 SECTION 24, T153N, R101W
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Drawn By: B.J.H. Project No: S14-09-254
 Checked By: O.O.K. Date: NOV, 2014

Revision No.	Date	By	Description



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Checked By: D.D.K. Date: NOV. 2014

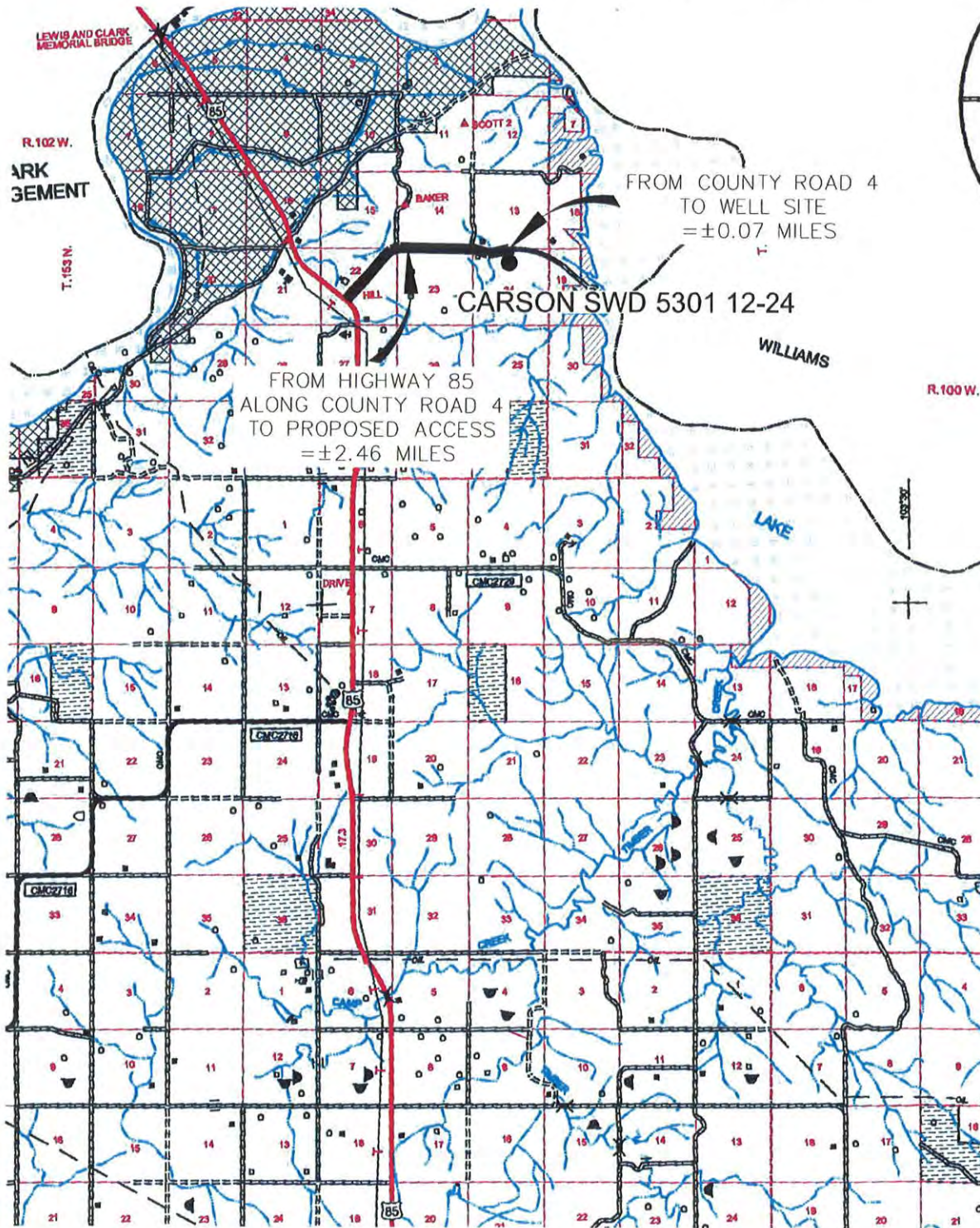
Revision No.	Date	By	Description

SHEET NO.

COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"CARSON SWD 5301 12-24"

584 FEET FROM NORTH LINE AND 2477 FEET FROM WEST LINE
SECTION 24, T153N, R101W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA



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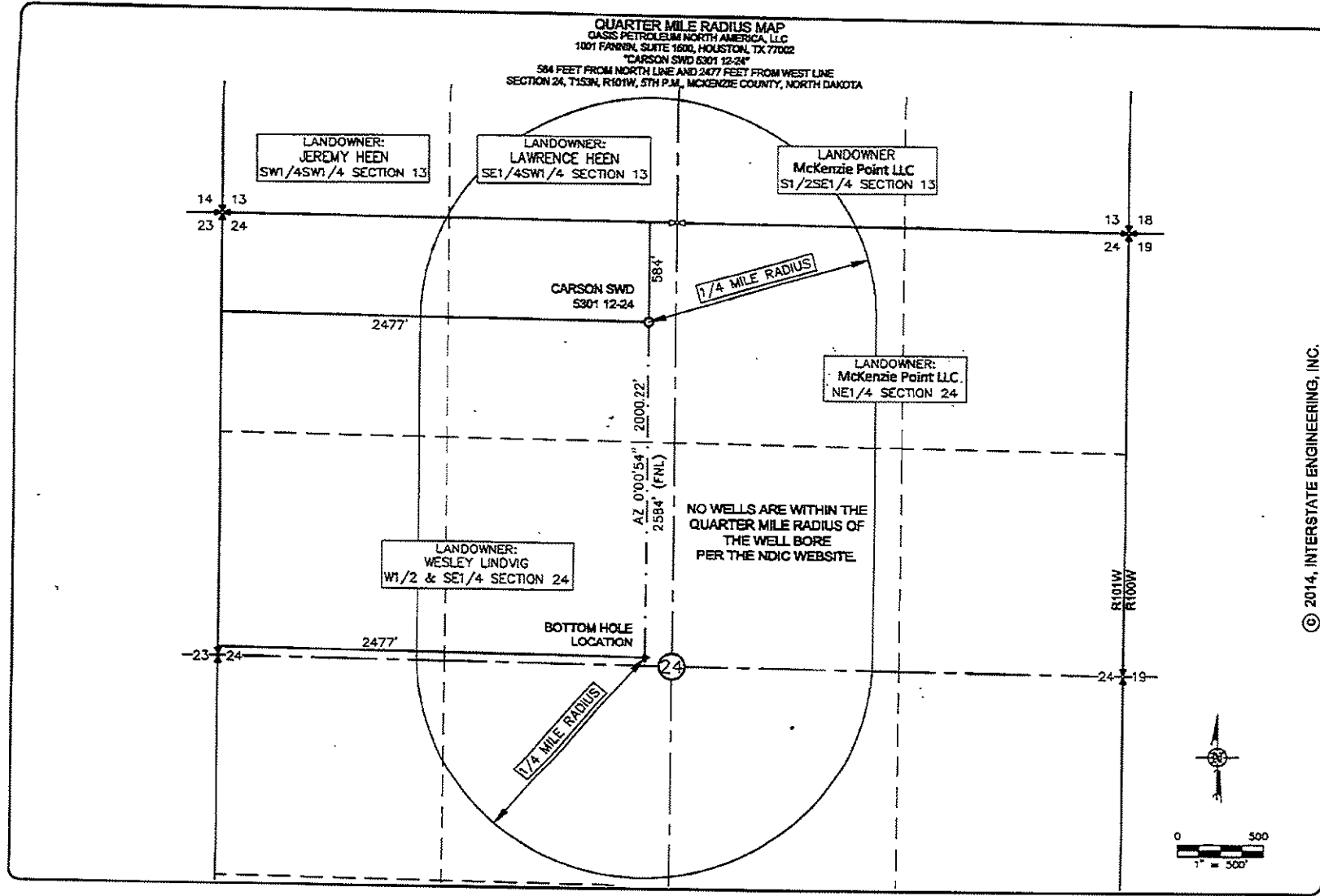
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COUNTY ROAD MAP
SECTION 24, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.J.H. Project No.: S14-09-254
Checked By: D.D.K. Date: NOV, 2014

Revision No.	Date	By	Description

SHEET NO.

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OASIS PETROLEUM NORTH AMERICA, LLC 1001 FANNIN, SUITE 1600 HOUSTON, TEXAS 77002 PHONE: (713) 433-8118 FAX: (713) 433-8119 WWW: OASIS-USA.COM		Project No.: Date: Drawn By: JES Checked By: JES
INTERSTATE ENGINEERING 423 East 10th Street Bismarck, ND 58101 Phone: (701) 339-8118 Fax: (701) 339-8119 Website: www.interstate-engineering.com		Project No.: Date: Drawn By: JES Checked By: JES

Area of Review Plat

15310113CC
Test Hole
Gregory Water & Energy Inc.-7/7/2009
Jeremy Heem

15310124AAA
Test Hole
S & S Drilling-5/7/1994
Tim Novak

15310124AAA
Test Hole
S & S Drilling-5/15/1994
Tim Novak

T153N
R101W

T153N
R100W

SHL

1/4 mile

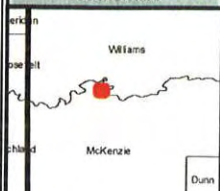
BHL

OASIS PETROLEUM
CHALMERS 5300 5300 31-19H
31-19H ST1

EARTHSTONE ENERGY
INCORPORATED
DAHL EDWIN M 23-1

Indian
Creek

LOCATION MAP



CARSON SWD 5301 12-24
153N - 101W, Section 24
McKenzie County, North Dakota

Carson SWD 5301 12-24
Oasis Petroleum North America, LLC
Section 24, T153N R101W
McKenzie County, North Dakota

Application for Fluid Injection

CORRECTIVE ACTIONS

A search of the NDIC ArcIMS system yielded no wells within a ¼ mile area of the proposed Carson SWD 5301 12-24. No corrective action is required.

AFFIDAVIT OF MAILING

STATE OF NORTH DAKOTA §
 §
COUNTY OF MCKENZIE §

The undersigned, Patrick McCauley, of lawful age, first duly sworn on his oath states that he is a duly authorized agent of Oasis Petroleum North America LLC, do hereby certify that I have sent (via certified mail) a Notice of Application for Fluid Injection to each surface owner within a radius of ¼ mile from the location of the proposed injection well (Carson SWD 5301 12-24), T153N-R101W, Section 24: NW4, N2SW4, McKenzie County, ND.

The following contains the names and addresses of all landowners and legal description of the land ownership:

Surface Owners

Lawrence P. Heen
4412 Highway 85 North
Williston, ND 58801

Jeremy A. Heen
4412 Highway 85 North
Williston, ND 58801

McKenzie Point, LLC
5009 139th Ave. NW
Williston, ND 58801

Wesley Lindvig and Barbara Lindvig
Tenants in Common
14075 41st Street NW
Alexander, ND 58831

Legal Description

Township 153 North, Range 101 West
Section 13: SE4SW4

Township 153 North, Range 101 West
Section 13: SW4SW4

Township 153 North, Range 101 West
Section 13: E2SE4, SW4SE4
Section 24: NE4

Township 153 North, Range 101 West
Section 24: NW4, N2SW4, NW4SE4

IN WITNESS WHEREOF, the undersigned has hereunto set his hand this 2nd day of December, 2014.

Oasis Petroleum North America LLC

By: Patrick McCauley
Patrick McCauley, Landman

ACKNOWLEDGMENT

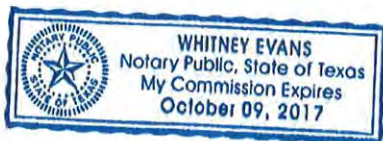
STATE OF TEXAS

§
§
§

COUNTY OF HARRIS

BE IT REMEMBERED, that on this 2nd day of December, 2014, before me, a Notary Public, in and for said County and State, personally appeared Patrick McCauley, to me known to be the identical person described in and who executed the within and foregoing instrument and acknowledged to me that he executed the same as his free and voluntary act and deed for the uses and purposes therein set forth.

IN WITNESS WHEREOF, I have hereto set my official signature and affixed my notarial seal, the day and year first above written.



Whitney Evans
Notary Public
My Commission Expires: 10.09.2017



IN WITNESS WHEREOF, the undersigned has hereunto set his hand this 7th day of January, 2015.

Oasis Petroleum North America LLC

By: Patrick McCauley
Patrick McCauley, Landman

ACKNOWLEDGMENT

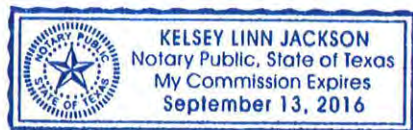
STATE OF TEXAS

§
§
§

COUNTY OF HARRIS

BE IT REMEMBERED, that on this 7th day of January, 2015, before me, a Notary Public, in and for said County and State, personally appeared Patrick McCauley, to me known to be the identical person described in and who executed the within and foregoing instrument and acknowledged to me that he executed the same as his free and voluntary act and deed for the uses and purposes therein set forth.

IN WITNESS WHEREOF, I have hereto set my official signature and affixed my notarial seal, the day and year first above written.



Kelsey Linn Jackson
Notary Public
My Commission Expires: 9-13-2016



SENDER: COMPLETE THIS SECTION		COMPLETE THIS SECTION ON DELIVERY	
■ Complete items 1, 2, and 3. Also complete item 4 if Restricted Delivery is desired. ■ Print your name and address on the reverse so that we can return the card to you. ■ Attach this card to the back of the mailpiece, or on the front if space permits.		A. Signature X <i>Heen</i> ☐ Agent ☐ Addressee	
1. Article Addressed to: Jeremy A. Heen 4502 139 th Ave. NW Alexander, ND 58831		B. Received by (Printed Name)	C. Date of Delivery
2. Article Number (Transfer from service label)		D. Is delivery address different from item 1? ☐ Yes If YES, enter delivery address below: ☐ No	
PS Form 3811, July 2013		3. Service Type ☒ Certified Mail® ☐ Priority Mail Express™ ☐ Registered ☒ Return Receipt for Merchandise ☐ Insured Mail ☐ Collect on Delivery	
		4. Restricted Delivery? (Extra Fee) ☐ Yes	
		7013 2250 0001 3657 6080	

Domestic Return Receipt

SENDER: COMPLETE THIS SECTION		COMPLETE THIS SECTION ON DELIVERY	
■ Complete items 1, 2, and 3. Also complete item 4 if Restricted Delivery is desired. ■ Print your name and address on the reverse so that we can return the card to you. ■ Attach this card to the back of the mailpiece, or on the front if space permits.		A. Signature X ☐ Agent ☐ Addressee	
1. Article Addressed to: Lawrence P. Heen 14033 45 th St. NW Alexander, ND 58831		B. Received by (Printed Name)	C. Date of Delivery
2. Article Number (Transfer from service label)		D. Is delivery address different from item 1? ☐ Yes If YES, enter delivery address below: ☐ No	
PS Form 3811, July 2013		3. Service Type ☒ Certified Mail® ☐ Priority Mail Express™ ☐ Registered ☒ Return Receipt for Merchandise ☐ Insured Mail ☐ Collect on Delivery	
		4. Restricted Delivery? (Extra Fee) ☐ Yes	
		7013 2250 0001 3657 6073	

Domestic Return Receipt

AFFIDAVIT OF MAILING

STATE OF NORTH DAKOTA

COUNTY OF MCKENZIE

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The undersigned, Patrick McCauley, of lawful age, first duly sworn on his oath states that he is a duly authorized agent of Oasis Petroleum North America LLC, do hereby certify that I have sent (via certified mail) a Notice of Application for Fluid Injection to each surface owner within a radius of ¼ mile from the location of the proposed injection well (Carson SWD 5301 12-24), T153N-R101W, Section 24: NW4, N2SW4, McKenzie County, ND.

The following contains the names and addresses of all landowners and legal description of the land ownership:

Surface Owners

Lawrence P. Heen
14033 45th St NW
Alexander, ND 58831

Jeremy A. Heen
4502 139th Ave. NW
Alexander, ND 58831

McKenzie Point, LLC
5009 139th Ave. NW
Williston, ND 58801

Wesley Lindvig and Barbara Lindvig
Tenants in Common
14075 41st Street NW
Alexander, ND 58831

Legal Description

Township 153 North, Range 101 West
Section 13: SE4SW4

Township 153 North, Range 101 West
Section 13: SW4SW4

Township 153 North, Range 101 West
Section 13: E2SE4, SW4SE4
Section 24: NE4

Township 153 North, Range 101 West
Section 24: NW4, N2SW4, NW4SE4



January 7, 2015

CERTIFIED MAIL
RETURN RECEIPT REQUESTED

Lawrence P. Heen
14033 45th St NW
Alexander, ND 58831

RE: Proposed Saltwater Disposal Well
Carson SWD 5301 12-24
T153N-R101W Section 24: NW4, N2SW4
McKenzie County, North Dakota

Dear Mr. Heen,

This letter is to advise you that Oasis Petroleum North America LLC is planning a saltwater disposal well and inject fluids (saltwater) from source wells in the area into the referenced location.

In accordance with the rules and regulations of the North Dakota Industrial Commission ("NDIC"), Oasis is required to give notice that it has made application to perform this work to each landowner within a one-quarter mile radius of the injection site. At a date yet to be determined, the NDIC will conduct a hearing regarding this application in the Department of Mineral Resources Conference Room located at 1000 East Calgary Avenue, Bismarck, ND 58503. Your comments or objections regarding this application may be directed to the Commission at that time.

Written comments or objections may be submitted prior to the hearing to the following address:

North Dakota Industrial Commission
600 E. Boulevard Avenue, Dept. 405
Bismarck, ND 58505

If you have any questions, concerns, or to determine the hearing date, please contact Kevin Connors, UIC/CCS Supervisor for the NDIC Oil & Gas Division at 701-328-8020.

Sincerely,

Patrick McCauley
Landman
(281) 404-9547
pmccauley@oasispetroleum.com



January 7, 2015

CERTIFIED MAIL
RETURN RECEIPT REQUESTED

Jeremy A. Heen
4502 139th Ave. NW
Alexander, ND 58831

RE: Proposed Saltwater Disposal Well
Carson SWD 5301 12-24
T153N-R101W Section 24: NW4, N2SW4
McKenzie County, North Dakota

Dear Mr. Heen,

This letter is to advise you that Oasis Petroleum North America LLC is planning a saltwater disposal well and inject fluids (saltwater) from source wells in the area into the referenced location.

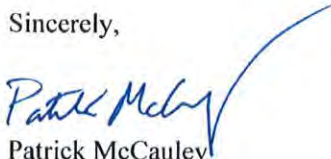
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Bismarck, ND 58505

If you have any questions, concerns, or to determine the hearing date, please contact Kevin Connors, UIC/CCS Supervisor for the NDIC Oil & Gas Division at 701-328-8020.

Sincerely,



Patrick McCauley

Landman
(281) 404-9547
pmccauley@oasispetroleum.com



December 2nd, 2014

CERTIFIED MAIL
RETURN RECEIPT REQUESTED

Lawrence P. Heen
4412 Highway 85 North
Williston, ND 58801

RE: Proposed Saltwater Disposal Well
Carson SWD 5301 12-24
T153N-R101W Section 24: NW4, N2SW4
McKenzie County, North Dakota

Dear Mr. Heen,

This letter is to advise you that Oasis Petroleum North America LLC is planning a saltwater disposal well and inject fluids (saltwater) from source wells in the area into the referenced location.

In accordance with the rules and regulations of the North Dakota Industrial Commission ("NDIC"), Oasis is required to give notice that it has made application to perform this work to each landowner within a one-quarter mile radius of the injection site. At a date yet to be determined, the NDIC will conduct a hearing regarding this application in the Department of Mineral Resources Conference Room located at 1000 East Calgary Avenue, Bismarck, ND 58503. Your comments or objections regarding this application may be directed to the Commission at that time.

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Bismarck, ND 58505

If you have any questions, concerns, or to determine the hearing date, please contact Kevin Connors, UIC/CCS Supervisor for the NDIC Oil & Gas Division at 701-328-8020.

Sincerely,

A handwritten signature in blue ink that reads "Patrick McCauley".

Patrick McCauley
Landman
(281) 404-9547
pmccauley@oasispetroleum.com



December 2nd, 2014

CERTIFIED MAIL
RETURN RECEIPT REQUESTED

Jeremy A. Heen
4412 Highway 85 North
Williston, ND 58801

RE: Proposed Saltwater Disposal Well
Carson SWD 5301 12-24
T153N-R101W Section 24: NW4, N2SW4
McKenzie County, North Dakota

Dear Mr. Heen,

This letter is to advise you that Oasis Petroleum North America LLC is planning a saltwater disposal well and inject fluids (saltwater) from source wells in the area into the referenced location.

In accordance with the rules and regulations of the North Dakota Industrial Commission ("NDIC"), Oasis is required to give notice that it has made application to perform this work to each landowner within a one-quarter mile radius of the injection site. At a date yet to be determined, the NDIC will conduct a hearing regarding this application in the Department of Mineral Resources Conference Room located at 1000 East Calgary Avenue, Bismarck, ND 58503. Your comments or objections regarding this application may be directed to the Commission at that time.

Written comments or objections may be submitted prior to the hearing to the following address:

North Dakota Industrial Commission
600 E. Boulevard Avenue, Dept. 405
Bismarck, ND 58505

If you have any questions, concerns, or to determine the hearing date, please contact Kevin Connors, UIC/CCS Supervisor for the NDIC Oil & Gas Division at 701-328-8020.

Sincerely,

A handwritten signature in blue ink that reads "Patrick McCauley".

Patrick McCauley
Landman
(281) 404-9547
pmccauley@oasispetroleum.com



December 2nd, 2014

CERTIFIED MAIL
RETURN RECEIPT REQUESTED

McKenzie Point, LLC
5009 139th Ave. NW
Williston, ND 58801

RE: Proposed Saltwater Disposal Well
Carson SWD 5301 12-24
T153N-R101W Section 24: NW4, N2SW4
McKenzie County, North Dakota

Dear McKenzie Point, LLC,

This letter is to advise you that Oasis Petroleum North America LLC is planning a saltwater disposal well and inject fluids (saltwater) from source wells in the area into the referenced location.

In accordance with the rules and regulations of the North Dakota Industrial Commission ("NDIC"), Oasis is required to give notice that it has made application to perform this work to each landowner within a one-quarter mile radius of the injection site. At a date yet to be determined, the NDIC will conduct a hearing regarding this application in the Department of Mineral Resources Conference Room located at 1000 East Calgary Avenue, Bismarck, ND 58503. Your comments or objections regarding this application may be directed to the Commission at that time.

Written comments or objections may be submitted prior to the hearing to the following address:

North Dakota Industrial Commission
600 E. Boulevard Avenue, Dept. 405
Bismarck, ND 58505

If you have any questions, concerns, or to determine the hearing date, please contact Kevin Connors, UIC/CCS Supervisor for the NDIC Oil & Gas Division at 701-328-8020.

Sincerely,

A handwritten signature in blue ink that reads "Patrick McCauley".

Patrick McCauley
Landman
(281) 404-9547
pmccauley@oasispetroleum.com



December 2nd, 2014

CERTIFIED MAIL
RETURN RECEIPT REQUESTED

Wesley Lindvig and Barbara Lindvig
Tenants in Common
14075 41st Street NW
Alexander, ND 58831

RE: Proposed Saltwater Disposal Well
Carson SWD 5301 12-24
T153N-R101W Section 24: NW4, N2SW4
McKenzie County, North Dakota

Dear Mr. and Mrs. Lindvig,

This letter is to advise you that Oasis Petroleum North America LLC is planning a saltwater disposal well and inject fluids (saltwater) from source wells in the area into the referenced location.

In accordance with the rules and regulations of the North Dakota Industrial Commission ("NDIC"), Oasis is required to give notice that it has made application to perform this work to each landowner within a one-quarter mile radius of the injection site. At a date yet to be determined, the NDIC will conduct a hearing regarding this application in the Department of Mineral Resources Conference Room located at 1000 East Calgary Avenue, Bismarck, ND 58503. Your comments or objections regarding this application may be directed to the Commission at that time.

Written comments or objections may be submitted prior to the hearing to the following address:

North Dakota Industrial Commission
600 E. Boulevard Avenue, Dept. 405
Bismarck, ND 58505

If you have any questions, concerns, or to determine the hearing date, please contact Kevin Connors, UIC/CCS Supervisor for the NDIC Oil & Gas Division at 701-328-8020.

Sincerely,

A handwritten signature in blue ink, appearing to read "Patrick McCauley".

Patrick McCauley
Landman
(281) 404-9547
pmccauley@oasispetroleum.com

AFFIDAVIT

STATE OF NORTH DAKOTA

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COUNTY OF MCKENZIE

The undersigned, Thomas Osen, of lawful age, first duly sworn on his oath states that he is a duly authorized agent of Oasis Petroleum North America LLC, do hereby certify that I have attempted to obtain water samples from the two nearest freshwater wells within a one-mile radius of the proposed location of the Carson SWD 5301 12-24, located in T153N-R101W, Section 24: NW4, N2SW4 in McKenzie County, ND.

I have obtained a water sample from a house spigot at a nearby residence that has been analyzed and will be submitted with the permit. I have diligently attempted to locate water wells within the area of review and have not been able to locate one. The one we have located does not have electricity and we are still attempting to acquire a sample from it.

IN WITNESS WHEREOF, the undersigned has hereunto set his hand this 3rd day of December, 2014.

Oasis Petroleum North America LLC

By:

Thomas Osen
Thomas Osen

ACKNOWLEDGMENT

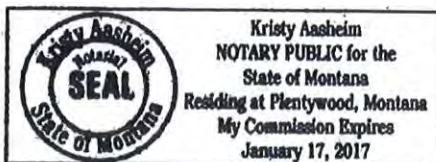
STATE OF North Dakota

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COUNTY OF Williams

BE IT REMEMBERED, that on this 3rd day of December, 2014, before me, a Notary Public, in and for said County and State, personally appeared **Thomas Osen**, to me known to be the identical person described in and who executed the within and foregoing instrument and acknowledged to me that he executed the same as his free and voluntary act and deed for the uses and purposes therein set forth.

IN WITNESS WHEREOF, I have hereto set my official signature and affixed my notarial seal, the day and year first above written.



Kristy Aasheim
Notary Public
My Commission Expires: 1/17/17

ASTRO-CHEM LAB, INC.

4102 2nd Ave. West

Williston, North Dakota 58802-0972
P.O. Box 972

Phone: (701) 572-7355

WATER ANALYSIS REPORT

Sample Number: W-14-9192

Date of Analysis: 11/05/2014

Company: Oasis Petroleum

City: Williston

State: ND

Well Number: Lindvig S.W.D.

Date Received: 10/30/2014

DST Number:

Sample Source: Brandon Turcotte Residence (House Spigot)

Location: SWSE

Section: 22

Township: 153

Range: 101

County:

Formation:

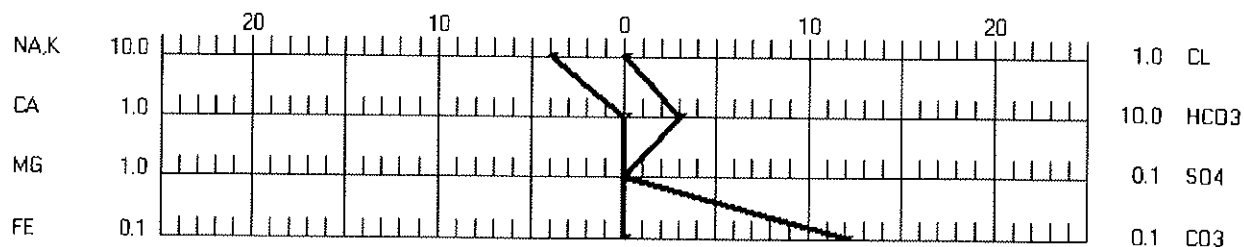
Depth:

Distribution: Thomas Osen

Resistivity @ 77 °F	3.540 Ohm-Meters	pH	8.35
Specific Gravity @ 77 °F	1.000	H2S	Negative
Total Dissolved Solids (Calculated)	2914 mg/L	(	2914 ppm)
Sodium Chloride (Calculated)	11 mg/L	(	11 ppm)

CATION	MEQ/L	mg/L	ANION	MEQ/L	mg/L
CALCIUM	0.2	4	CHLORIDE	0.2	7
MAGNESIUM	0.2	2	CARBONATE	1.2	36
SODIUM	37.4	860	BICARBONATE	32.8	2001
IRON	0.0	0.0	SULFATE	0.0	0
CHROMIUM	0.0	0.0	NITRATE-N	0.0	0
BARIUM	0.0	0.0			
POTASSIUM	0.1	4			

WATER ANALYSIS PATTERN



Remarks: Conductivity = 2822 μ mhos/cm / Sampled 10-28-14
Lat 48-03-12.4 / Long 103-29-25.18

Analyzed By: C. Jungels

ASTRO-CHEM LAB, INC.

4102 2nd Ave. West

Williston, North Dakota 58802-0972
P.O. Box 972

Phone: (701) 572-7355

WATER ANALYSIS REPORT

SAMPLE NUMBER W-12-2853 DATE OF ANALYSIS 4-25-12
COMPANY Oasis Petroleum
CITY Williston STATE ND
WELL NAME AND/OR NUMBER Chalmers 5300-31-19H
DATE RECEIVED 4-11-12 DST NUMBER
SAMPLE SOURCE
LOCATION Lot 3 OF SEC. 19 TWN. 153N RANGE 100W COUNTY McKenzie
FORMATION DEPTH
DISTRIBUTION Distribution List

RESISTIVITY @ 77°F = 0.038 Ohm-Meters pH = 5.47

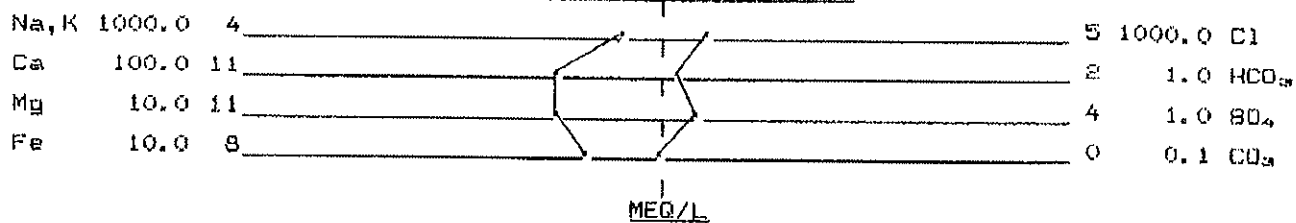
SPECIFIC GRAVITY @ 77°F = 1.215 H₂S = Negative

TOTAL DISSOLVED SOLIDS (CALCULATED) = 313035 mg/L (257642 ppm)

SODIUM CHLORIDE (CALCULATED) = 309798 mg/L (254978 ppm)

CATION	MEQ/L	mg/L	ANION	MEQ/L	mg/L
CALCIUM	1070.0	21229	CHLORIDE	5299.2	187873
MAGNESIUM	110.0	1222	CARBONATE	0.0	0
SODIUM	4045.3	93000	BICARBONATE	2.4	146
IRON	81.7	1520.0	SULFATE	3.8	182
CHROMIUM	0.1	1.0	NITRATE	0.0	0
BARIUM	0.6	41.7			
POTASSIUM	200.0	7820			

WATER ANALYSIS PATTERN



REMARKS Sampled 4-11-12

ANALYZED BY: C. Jungels

Preliminary List of Source Wells

Oasis Petroleum North America LLC
Application for Salt Water Disposal
Sec 24-153N-101W, McKenzie County, ND
Carson SWD 5301 12-24

Preliminary List of Source Wells*

Well Name: Chalmers 5300 31-19H
Operator: Oasis Petroleum North America LLC
Location: NWSW 19-153-100
API Number: 3305303472
NDIC Well File: 20407

Well Name: Chalmers 5300 21-19 10B
Operator: Oasis Petroleum North America LLC
Location: SWNW 19-153-100
API Number: 3305306022
NDIC Well File: 28637

Well Name: Kline Federal 5300 11-18 4T2
Operator: Oasis Petroleum North America LLC
Location: NWNW 18-153-100
API Number: 3305306224
NDIC Well File: 29243

Well Name: Kline Federal 5300 11-18 5B
Operator: Oasis Petroleum North America LLC
Location: NWNW 18-153-100
API Number: 3305306223
NDIC Well File: 29242

Well Name: Kline Federal 5300 11-18 6B
Operator: Oasis Petroleum North America LLC
Location: NWSW 18-153-100
API Number: 3305306057
NDIC Well File: 28756

**Subject to addition of operated and non-operated source wells.*

Surface Facilities Diagram

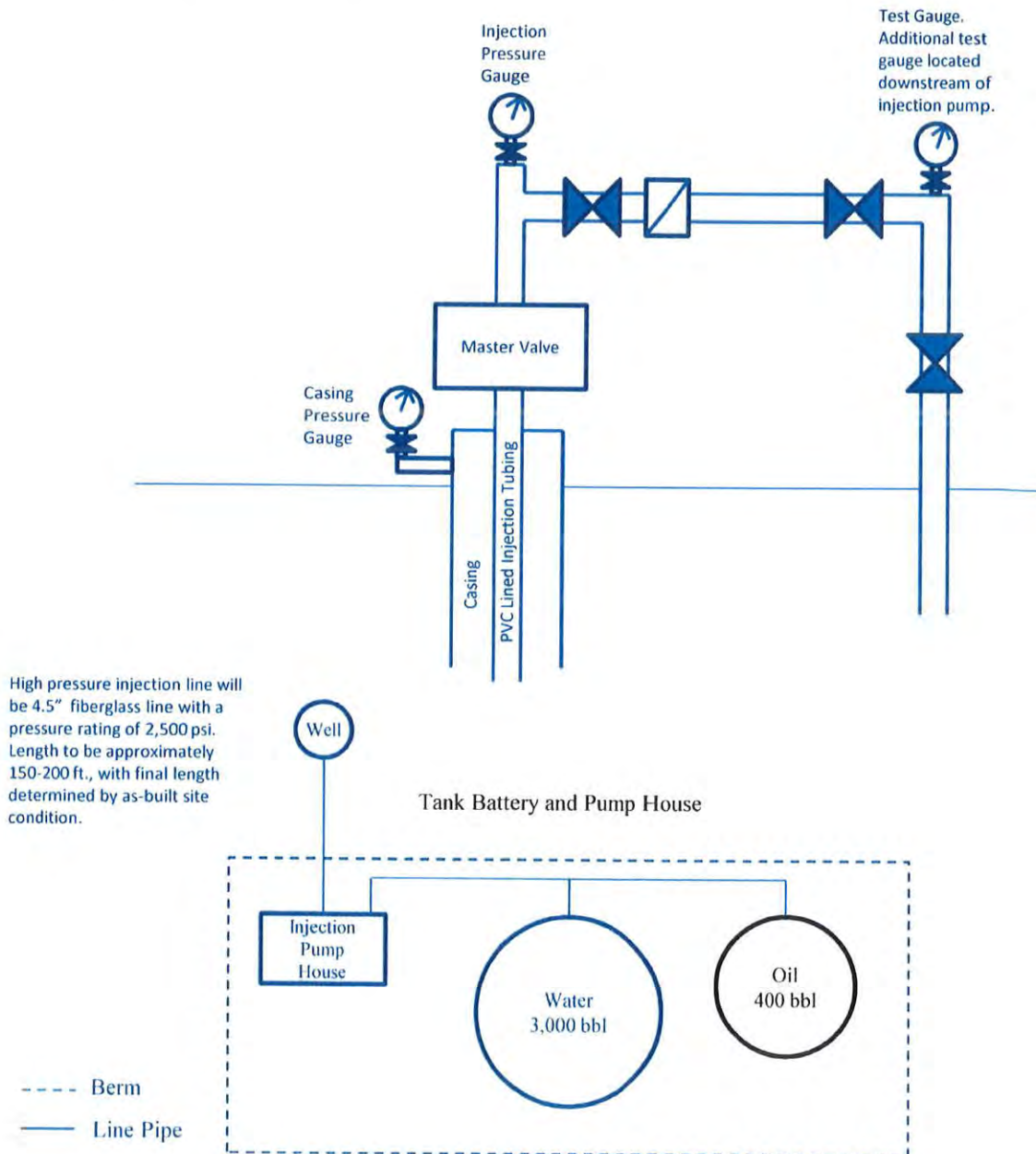
Surface Facilities Schematics

Carson SWD 5301 12-24

Oasis Petroleum

Section 24, T153N R101W

McKenzie County, North Dakota



Surface Facilities Diagram

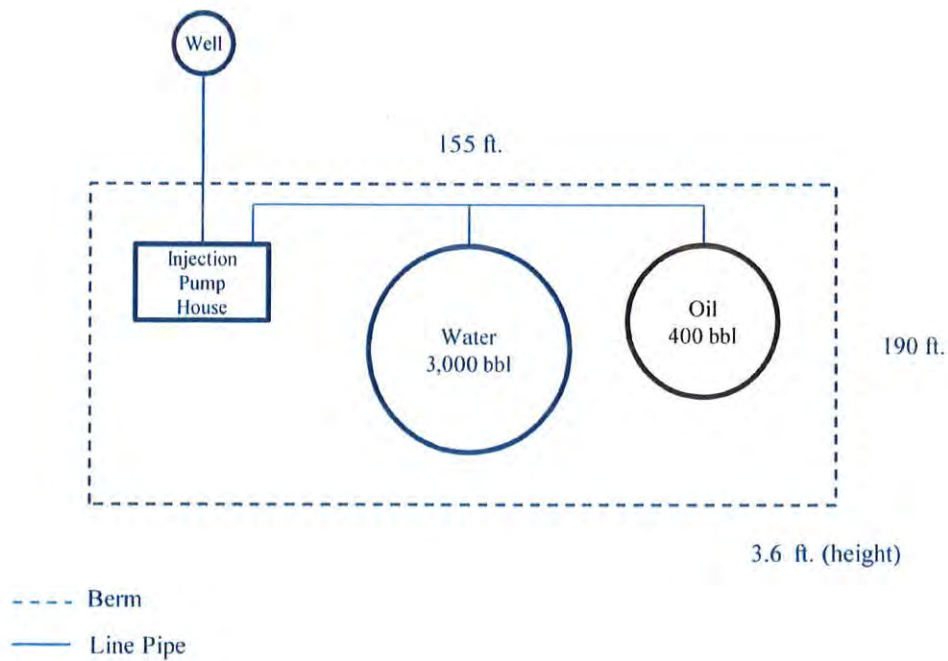
SWD Dike Schematics

Carson SWD 5301 12-24

Oasis Petroleum

Section 24, T153N R101W

McKenzie County, North Dakota



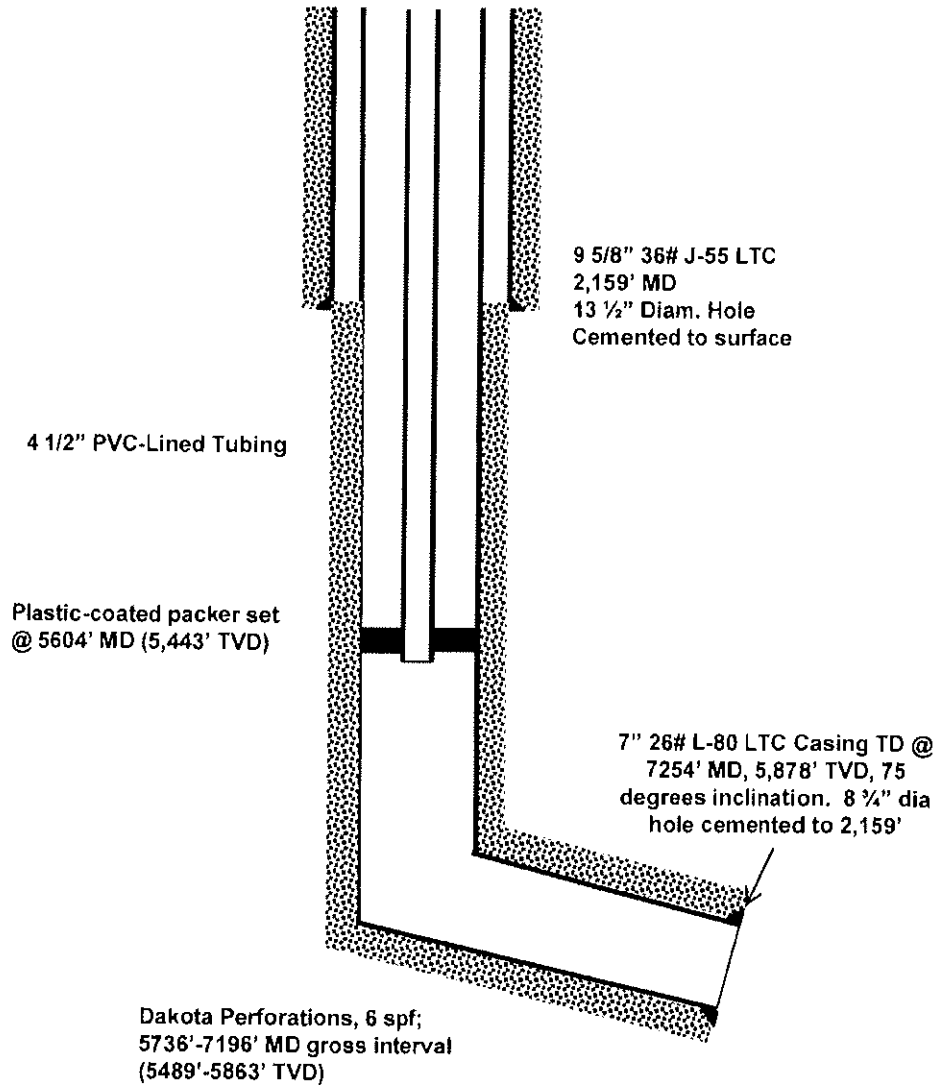
Dike Containment = 19,231 bbls

Pump House & Tanks Volume = 402 bbls

Effective Dike Containment = 18,829 bbls

**CARSON SWD 5301 12-24
WELLBORE SCHEMATIC**

API #:
FORMATION: Dakota
FIELD: Camp



OASIS PETROLEUM INC.

CARSON SWD 5301 12-24
T153N-R101W Sec. 24
584' FNL & 2477' FWL

Status: Planned
McKenzie County, North Dakota
Updated: 11/25/2014

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2008)

Well File No.

90329

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date February 1, 2015
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Proposed Procedure

Well Name and Number Carson SWD 5301 12-14 24					
Footages	Qtr-Qtr	Section	Township	Range	
584 F N L 2477 F W L	NENW	24	153 N	101 W	
Field Baker	Pool Dakota	County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Drill Procedure: 1. Notify state inspector prior to spud 2. Drill 13 1/2" hole to 2159' 3. TOOH and run 9 5/8" casing 3. Drill 8.75" hole to 4797' 4. Kickoff due South with BR's of 8deg/100' 4. Hold @ 75.15 deg inc to TD 6. TOOH and run 7" casing to bottom Completion Procedure: 1. Rig up pump truck and test wellbore to 1,500 psi. 2. MIRU wireline and mast truck. 3. RIH with perf assembly and perforate Dakota at 6 spf. 4. RU pump truck and perform injectivity test. If necessary, acidize perforations and add additional perfs. 5. PU plastic lined packer and new 4.5" plastic coated tubing. TIH and set packer 50' above top perforation. 6. Pressure test backside to 1,000 psi and record - MIT must be witnessed by a NDIC field inspector. 7. RD workover rig and install SWD facilities	<i>* Notify field inspector prior to beginning well site construction (KCC)</i>
--	---

Company Oasis Petroleum North America LLC	Telephone Number 281-404-9652
Address 1001 Fannin Suite 1500	
City Houston	State TX
Zip Code 77002	
Signature <i>[Signature]</i>	Printed Name Victoria Siemieniewski
Title Regulatory Specialist	Date December 8, 2014
Email Address vsiemieniewski@oasispetroleum.com	

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 3/20/2015	
By <i>[Signature]</i>	
Title KEVIN CONNORS UNDERGROUND INJECTION SUPERVISOR	



Oil and Gas Division

Lynn D. Helms - Director Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.oilgas.nd.gov

March 20, 2015

Victoria Siemieniewski
Regulatory Specialist
OASIS PETROLEUM NORTH AMERICA LLC
1001 Fannin, Suite 1500
Houston, TX 77002

RE: **DIRECTIONAL WELL**
 CARSON SWD 5301 12-24
 NENW Section 24-153N-101W
 McKenzie County
 Well File # 90329

Dear Victoria:

Pursuant to Commission Order No. 25893, approval to drill the above captioned well is hereby given.

PERMIT STIPULATIONS: OASIS PETROLEUM NORTH AMERICA must contact NDIC Field Inspector Richard Dunn at (701) 770-3554 prior to location construction.

Effective June 1, 2014, all operators of disposal wells in North Dakota are required to maintain an on-site container to store filters until they can be properly disposed of in an authorized facility. Such containers must be:

- **Leak-proof to prevent any fluids from escaping**
- **Covered to prevent precipitation from entering the container**
- **Labeled to indicate only filters are to be placed in the container**

Drilling pit

NDAC 43-02-03-19.4 states that "a pit may be utilized to bury drill cuttings and solids generated during well drilling and completion operations, providing the pit can be constructed, used and reclaimed in a manner that will prevent pollution of the land surface and freshwaters. Reserve and circulation of mud system through earthen pits are prohibited. All pits shall be inspected by an authorized representative of the director prior to lining and use. Drill cuttings and solids must be stabilized in a manner approved by the director prior to placement in a cuttings pit."

Form 1 Changes

Any changes, shortening of casing point or lengthening at Total Depth must have prior approval by the NDIC.

Permit Fee & Notification

Payment was received in the amount of \$100 via credit card. The permit fee has been received. It is requested that notification be given immediately upon the spudding of the well. This information should be relayed to the Oil & Gas Division, Bismarck, via telephone. The following information must be included: Well name, legal location, permit number, drilling contractor, company representative, date and time of spudding. Office hours are 8:00 a.m. to 12:00 p.m. and 1:00 p.m. to 5:00 p.m. Central Time. Our telephone number is (701) 328-8020, leave a message if after hours or on the weekend.

Survey Requirements for Horizontal, Horizontal Re-entry, and Directional Wells

NDAC Section 43-02-03-25 (Deviation Tests and Directional Surveys) states in part (that) the survey contractor shall file a certified copy of all surveys with the director free of charge within thirty days of completion. Surveys must be submitted as one electronic copy, or in a form approved by the director. However, the director may require the directional survey to be filed immediately after completion if the survey is needed to conduct the operation of the director's office in a timely manner. Certified surveys must be submitted via email in one adobe document, with a certification cover page to certsurvey@nd.gov. Survey points shall be of such frequency to accurately determine the entire location of the well bore. Specifically, the Horizontal and Directional well survey frequency is 100 feet in the vertical, 30 feet in the curve (or when sliding) and 90 feet in the lateral.

Conductors, Rat holes, and Mouse holes

To protect near surface groundwater any conductor, rat, or mouse hole drilled must be constructed with a string of casing and cemented to ground level. Any such string must be secured at the surface when not in use. In addition, all rat and mouse holes must be plugged with cement and cut off at least 4' below final grade within a reasonable timeframe after the rig completes drilling operations on the pad.

Surface casing cement

Tail cement utilized on surface casing must have a minimum compressive strength of 500 psi within 12 hours, and tail cement utilized on production casing must have a minimum compressive strength of 500 psi before drilling the plug or initiating tests.

Logs

NDAC Section 43-02-03-31 requires the running of (1) a suite of open hole logs from which formation tops and porosity zones can be determined, (2) a Gamma Ray Log run from total depth to ground level elevation of the well bore, and (3) a log from which the presence and quality of cement can be determined (Standard CBL or Ultrasonic cement evaluation log) in every well in which production or intermediate casing has been set, this log must be run prior to completing the well. All logs run must be submitted free of charge, as one digital TIFF (tagged image file format) copy and one digital LAS (log ASCII) formatted copy. Digital logs may be submitted on a standard CD, DVD, or attached to an email sent to digitallogs@nd.gov

Thank you for your cooperation.

Sincerely,

Kevin Connors
UIC\CCS Supervisor



APPLICATION FOR PERMIT TO DRILL - FORM 1

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 4615 (03-2006)

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.

PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Type of Work New Location	Type of Well Salt Water Disposal	Approximate Date Work Will Start 1 / 1 / 2015	Confidential Status No
Operator OASIS PETROLEUM NORTH AMERICA LLC		Telephone Number (281) 404-9500	
Address 1001 Fannin, Suite 1500		City Houston	State TX
Name of Surface Owner or Tenant Wesley and Barbara Lindvig		Zip Code 77002	
Address 1407 41st ST NW		City Alexander	State ND
		Zip Code 58831	

WELL INFORMATION

☒ Notice has been provided to the owner of any permanently occupied dwelling within 1,320 feet.

☒ This well is not located within five hundred feet of an occupied dwelling.

Well Name CARSON SWD				Well Number 5301 12-24			
At Surface 584 F N L 2477 F W L	Qtr-Qtr NENW	Section 24	Township 153 N	Range 101 W	County McKenzie		
If Directional, Top of Pay 1117 F N L 2477 F W L	Qtr-Qtr NENW	Section 24	Township 153 N	Range 101 W	County McKenzie		
Proposed Bottom Hole Location 2584 F N L 2477 F W L	Qtr-Qtr SWSE	Section 24	Township 153 N	Range 101 W	County McKenzie		
Latitude of Well Head 48° 03' 57.89"	Longitude of Well Head -103° 36' 56.05"	NAD Reference NAD83	Description of (Subject to NDIC Approval) Spacing Unit: Section 24 T153N R101W				
Ground Elevation 2087 Feet Above S.L.	Acres in Spacing/Drilling Unit	Spacing/Drilling Unit Setback Requirement Feet		Industrial Commission Order 25893			
Objective Horizons Dakota				Pierre Shale Top 2059			
Proposed Surface Casing 9 - 5/8"	Size 36 Lb./Ft.	Weight 2159 Feet	Cement Volume 635 Sacks	NOTE: Surface hole must be drilled with fresh water and surface casing must be cemented back to surface.			
Proposed Longstring Casing 7 - "	Size 26 Lb./Ft.	Weight(s) 7254 Feet MD	Longstring Total Depth 5878 Feet TVD	Cement Volume 509 Sacks	Cement Top 2159 Feet	Top Dakota Sand 5421 Feet	
Base Last Charles Salt (If Applicable) Feet	Estimated Total Depth (feet) 7254 Feet MD		5878 Feet TVD		Drilling Mud Type (Vertical Hole - Below Surface Casing) Invert		
Proposed Logs triple combo through mowry, GR/RES to BSC, GR to surf							

Comments

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.

Date

12 / 10 / 2014

ePermit

Printed Name

Victoria Siemieniowski

Title

Regulatory Specialist

FOR STATE USE ONLY

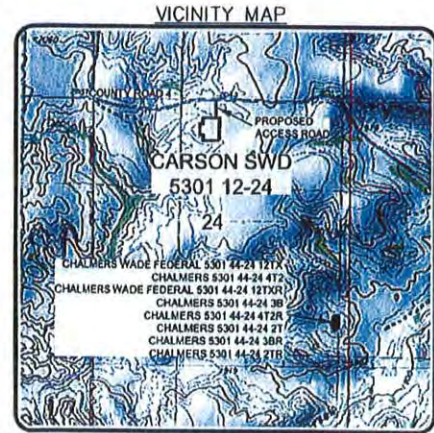
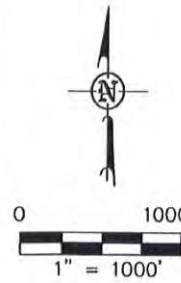
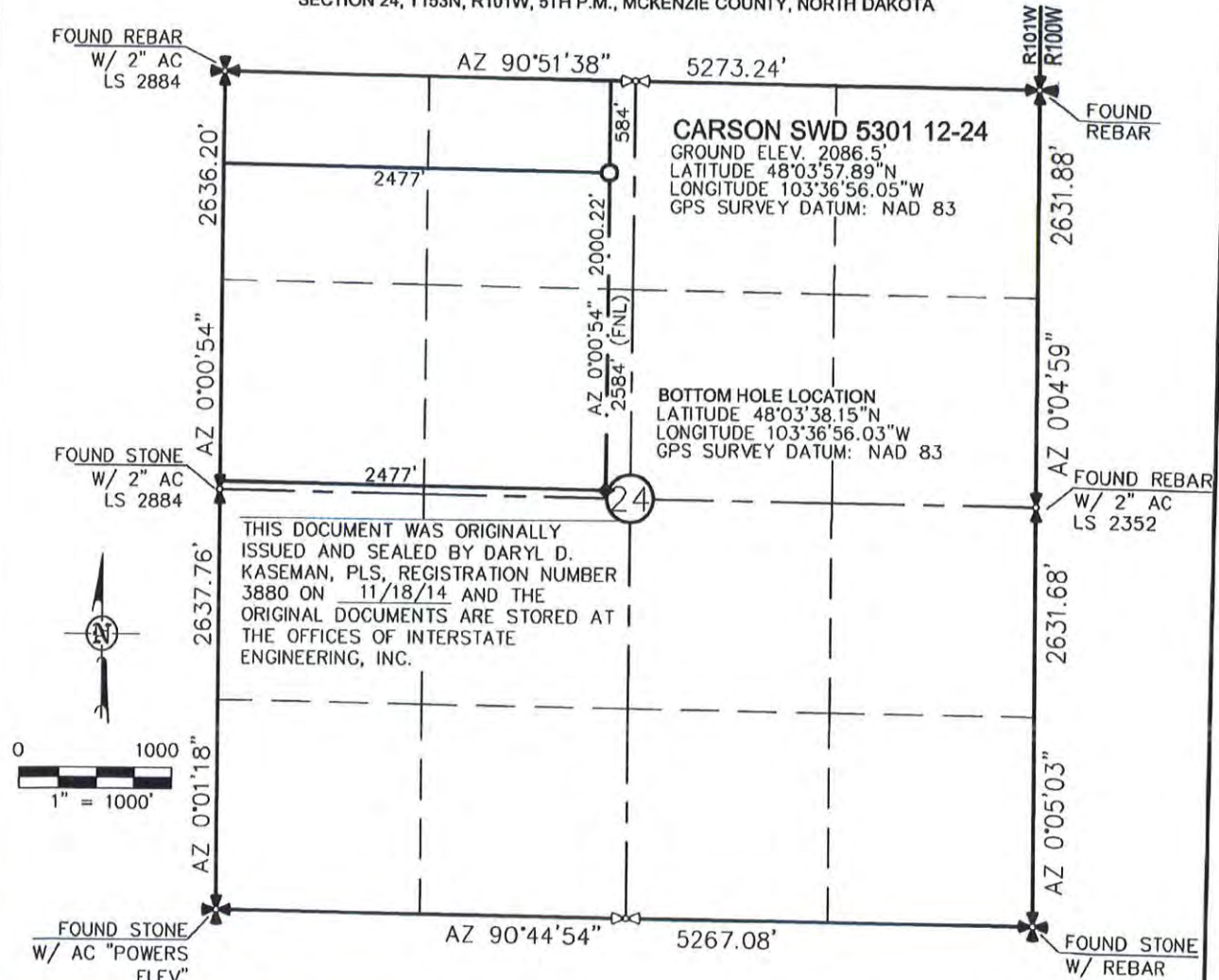
Permit and File Number 90329	API Number 33 - 053 - 90329
Field BAKER	
Pool DAKOTA	Permit Type SALT WATER DISPOSAL

FOR STATE USE ONLY

Date Approved 3 / 20 / 2015
By Kevin Connors
Title UICCCS Supervisor

REQUIRED ATTACHMENTS: Certified surveyors plat, estimated geological tops, proposed mud/cementing plans, \$100 fee.

WELL LOCATION PLAT
OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"CARSON SWD 5301 12-24"
 584 FEET FROM NORTH LINE AND 2477 FEET FROM WEST LINE
 SECTION 24, T153N, R101W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA



 - MONUMENT - RECOVERED
 - MONUMENT - NOT RECOVERED

STAKED ON 11/18/14
 VERTICAL CONTROL DATUM WAS BASED UPON
 CONTROL POINT 130856 WITH AN ELEVATION OF 2089.7'

THIS SURVEY AND PLAT IS BEING PROVIDED AT
 THE REQUEST OF ERIC BAYES OF OASIS
 PETROLEUM. I CERTIFY THAT THIS PLAT CORRECTLY
 REPRESENTS WORK PERFORMED BY ME OR UNDER
 MY SUPERVISION AND IS TRUE AND CORRECT TO
 THE BEST OF MY KNOWLEDGE AND BELIEF.

[Signature]

RECEIVED LAND SURVEYOR
DARYL D. KASEMAN
 LS-3880
 DATE: 11/18/14
 NORTH DAKOTA

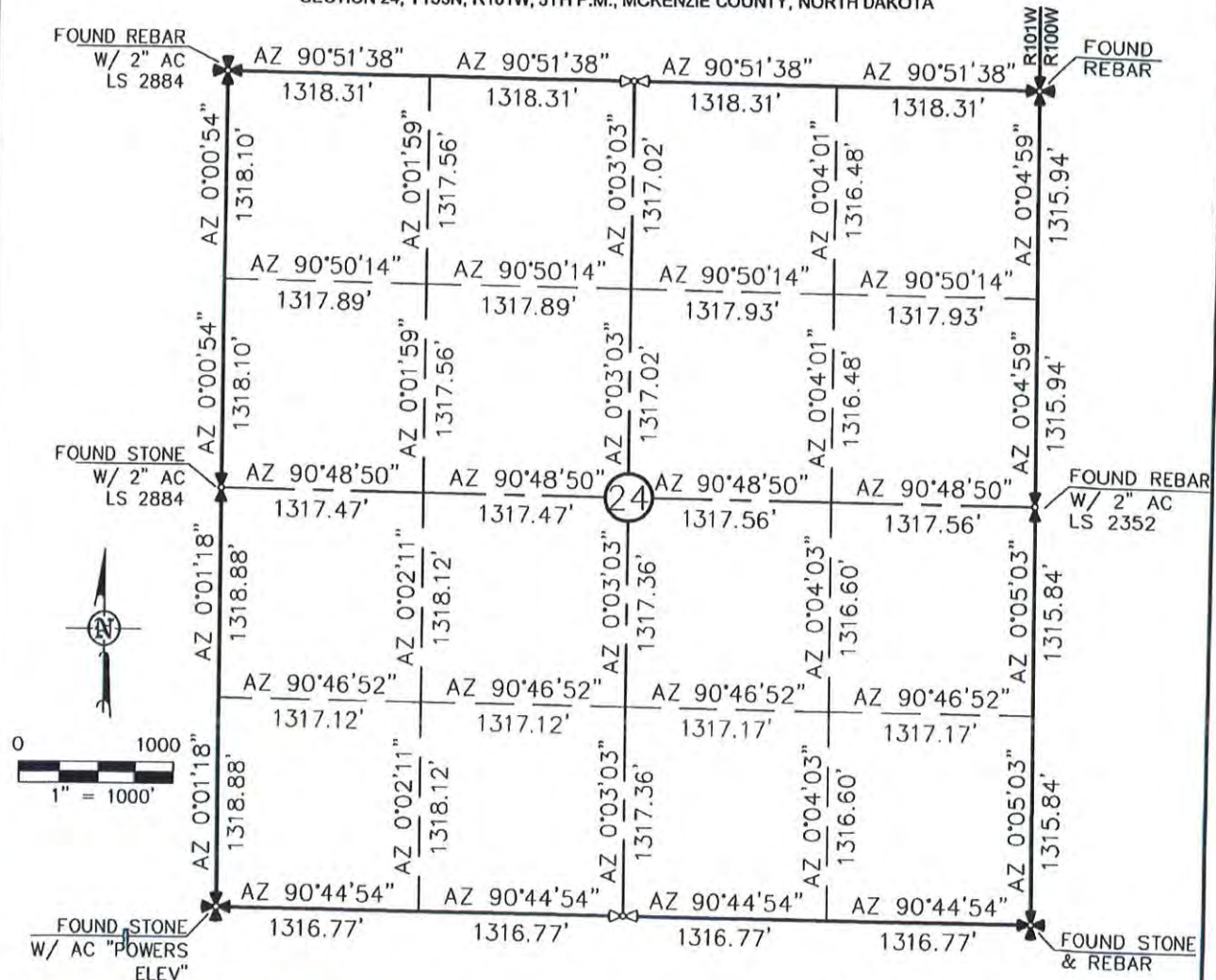
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DARYL D. KASEMAN LS-3880

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	Revision No.	Date	By	Description																			
SHEET NO.	4	4	4																				

SECTION BREAKDOWN
OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

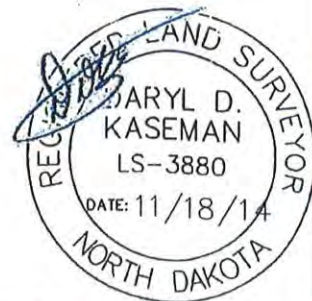
"CARSON SWD 5301 12-24"
584 FEET FROM NORTH LINE AND 2477 FEET FROM WEST LINE
SECTION 24, T153N, R101W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA



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ALL AZIMUTHS ARE BASED ON G.P.S. OBSERVATIONS. THE ORIGINAL SURVEY OF THIS AREA FOR THE GENERAL LAND OFFICE (G.L.O.) WAS 1901. THE CORNERS FOUND ARE AS INDICATED AND ALL OTHERS ARE COMPUTED FROM THOSE CORNERS FOUND AND BASED ON G.L.O. DATA. THE MAPPING ANGLE FOR THIS AREA IS APPROXIMATELY -0°03'

⛶ - MONUMENT - RECOVERED
⛶ - MONUMENT - NOT RECOVERED



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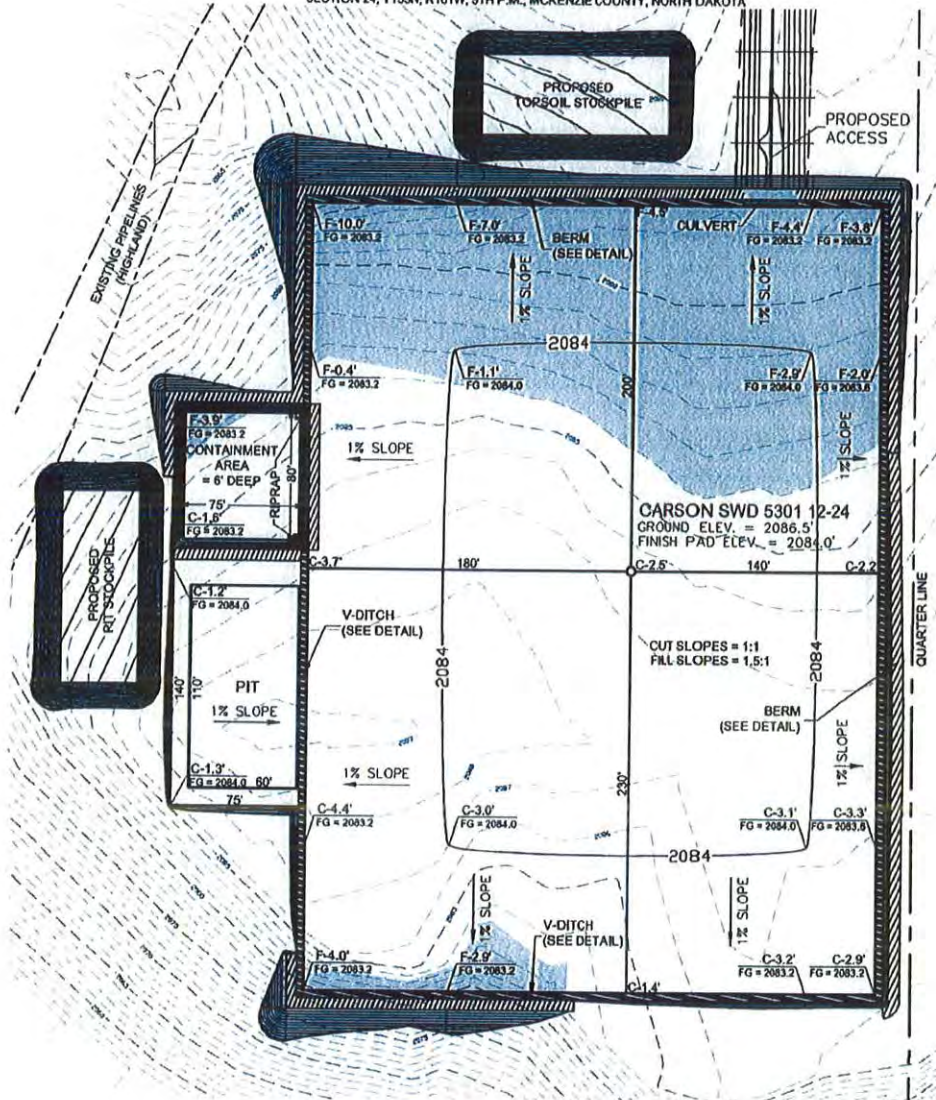
Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.interstateeng.com
Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
SECTION BREAKDOWN
SECTION 24, T153N, R101W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S14-09-254
Checked By: D.O.K. Date: NOV. 2014

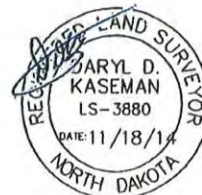
Revision No.	Date	By	Description

PAD LAYOUT
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "CARSON SWD 5301 12-24"
 584 FEET FROM NORTH LINE AND 2477 FEET FROM WEST LINE
 SECTION 24, T153N, R101W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA

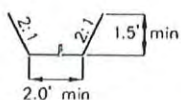


NOTE: Pad dimensions shown are to usable area, the v-ditch and berm areas shall be built to the outside of the pad dimensions.

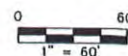
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V-DITCH DETAIL



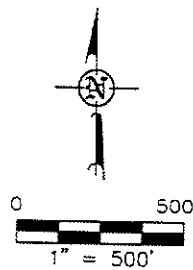
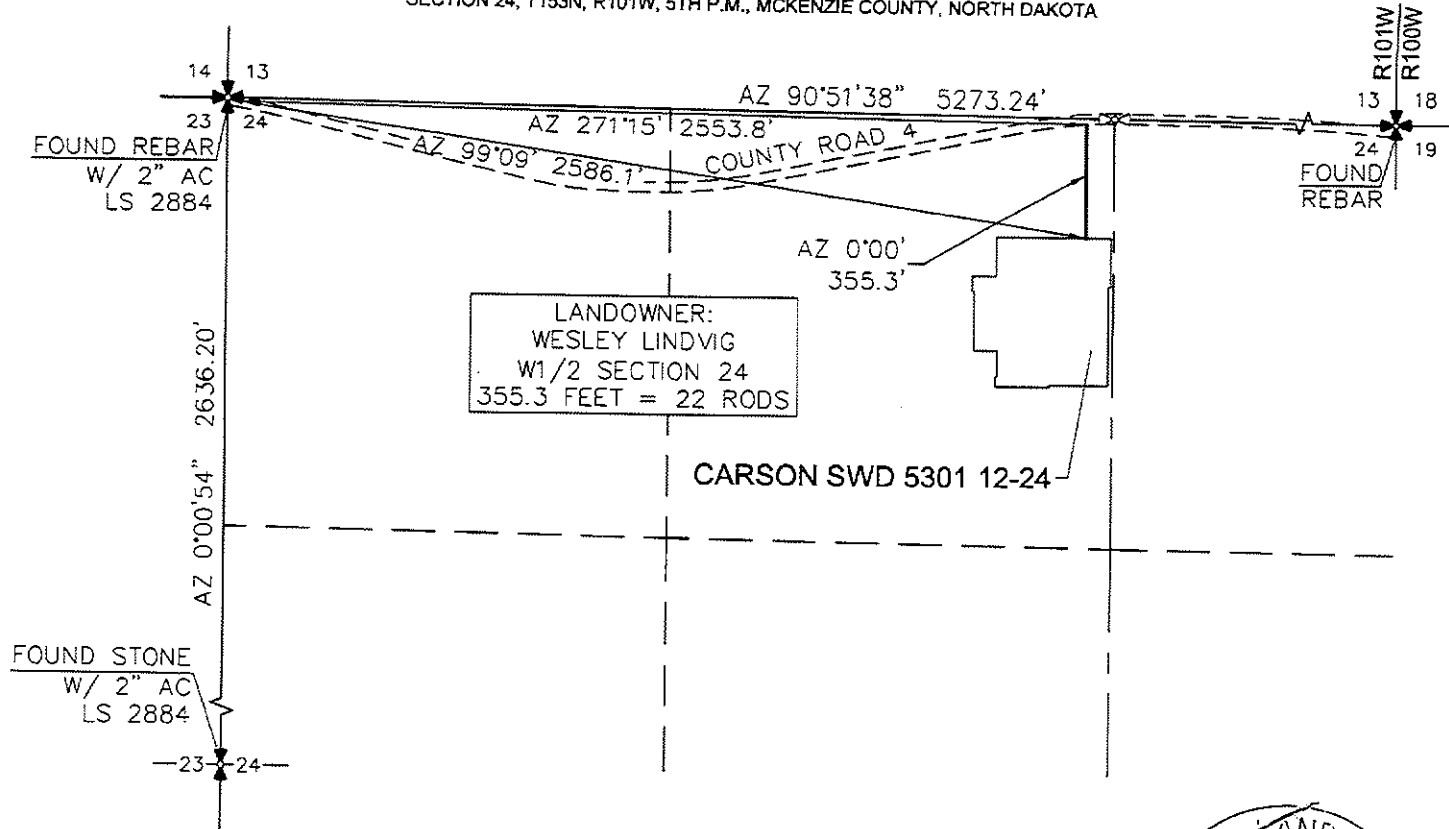
Proposed Contours
 Original Contours



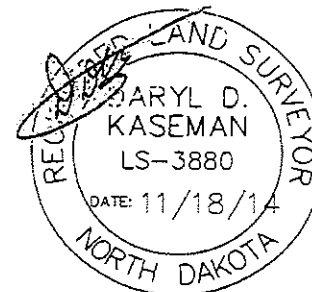
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Drawn By: <u>B.H.H.</u> Project No: <u>51409-754</u> Checked By: <u>D.D.K.</u> Date: <u>NOV. 2014</u>	SHEET NO.																		

ACCESS APPROACH
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "CARSON SWD 5301 12-24"
 584 FEET FROM NORTH LINE AND 2477 FEET FROM WEST LINE
 SECTION 24, T153N, R101W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA



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NOTE: All utilities shown are preliminary only, a complete
 utilities location is recommended before construction.

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OASIS PETROLEUM NORTH AMERICA, LLC ACCESS APPROACH SECTION 24, T153N, R101W MCKENZIE COUNTY, NORTH DAKOTA Drawn By: B.H.L. Project No.: S140234 Checked By: D.B.K. Date: NOV. 2014	

WELL LOCATION SITE QUANTITIES
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "CARSON SWD 5301 12-24"
 584 FEET FROM NORTH LINE AND 2477 FEET FROM WEST LINE
 SECTION 24, T153N, R101W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA

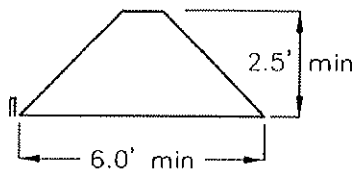
WELL SITE ELEVATION 2086.5
 WELL PAD ELEVATION 2084.0

EXCAVATION	12,315
PLUS PIT	3,150
ACCESS ROAD (CUT)	27
	<u>15,492</u>
EMBANKMENT	6,588
PLUS SHRINKAGE (30%)	1,976
ACCESS ROAD (FILL)	1,441
	<u>10,005</u>
STOCKPILE PIT	3,150
STOCKPILE TOP SOIL (6")	3,243
BERMS	1,332 LF = 432 CY
DITCHES	1,420 LF = 217 CY
CONTAINMENT AREA	1,121 CY
STOCKPILE MATERIAL	0
DISTURBED AREA FROM PAD	4.02 ACRES

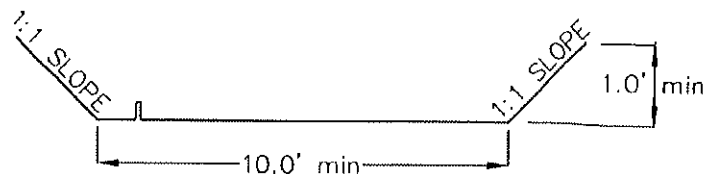
NOTE: ALL QUANTITIES ARE IN CUBIC YARDS (UNLESS NOTED)
 CUT END SLOPES AT 1:1
 FILL END SLOPES AT 1.5:1

WELL SITE LOCATION
 584' FNL
 2477' FWL

BERM DETAIL



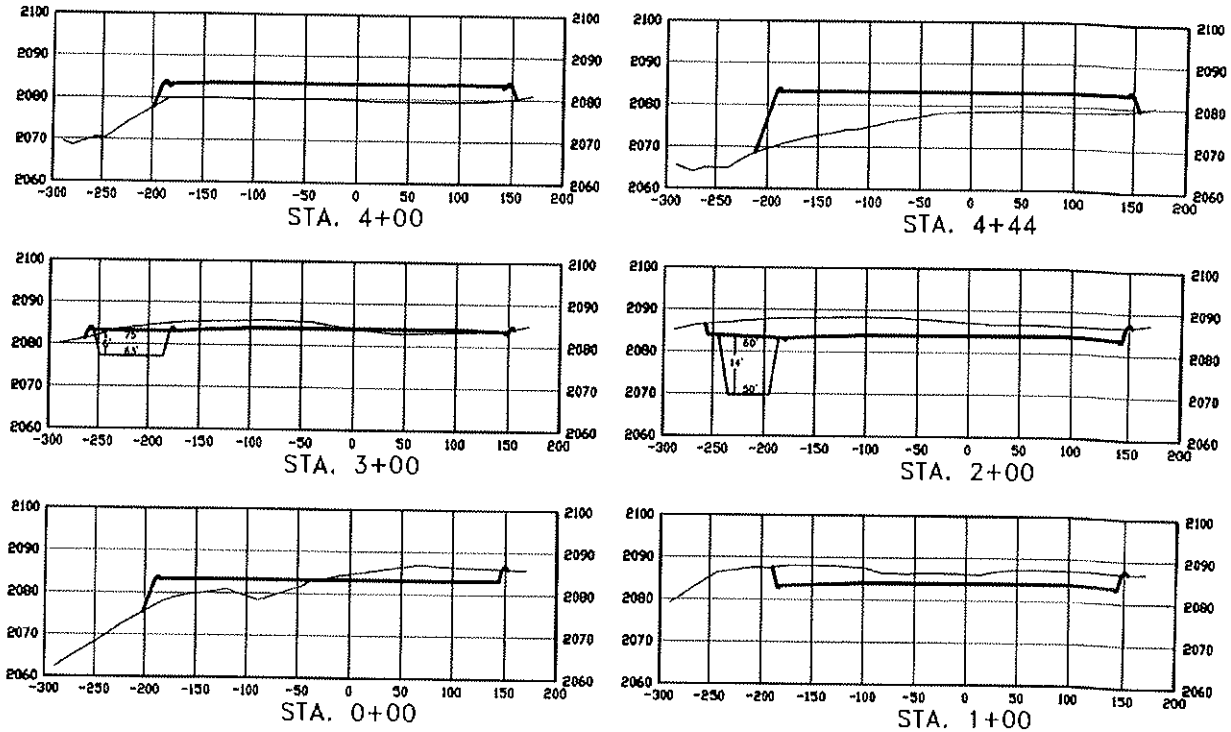
DIVERSION DITCH DETAIL



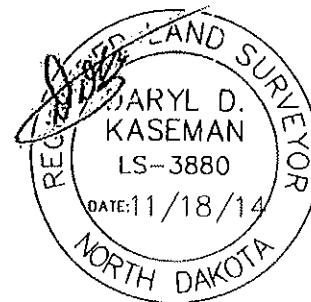
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			Drawn By: <u>BJH</u> Checked By: <u>DDK</u>	Project No.: <u>511-09-254</u> Date: <u>NOV. 2014</u>				

CROSS SECTIONS
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "CARSON SWD 5301 12-24"
 584 FEET FROM NORTH LINE AND 2477 FEET FROM WEST LINE
 SECTION 24, T153N, R101W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA



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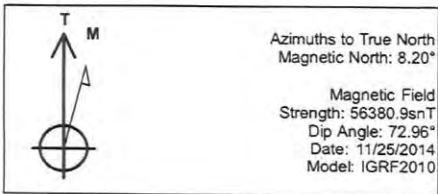


SCALE
 HORIZ 1"=160'
 VERT 1"=40'

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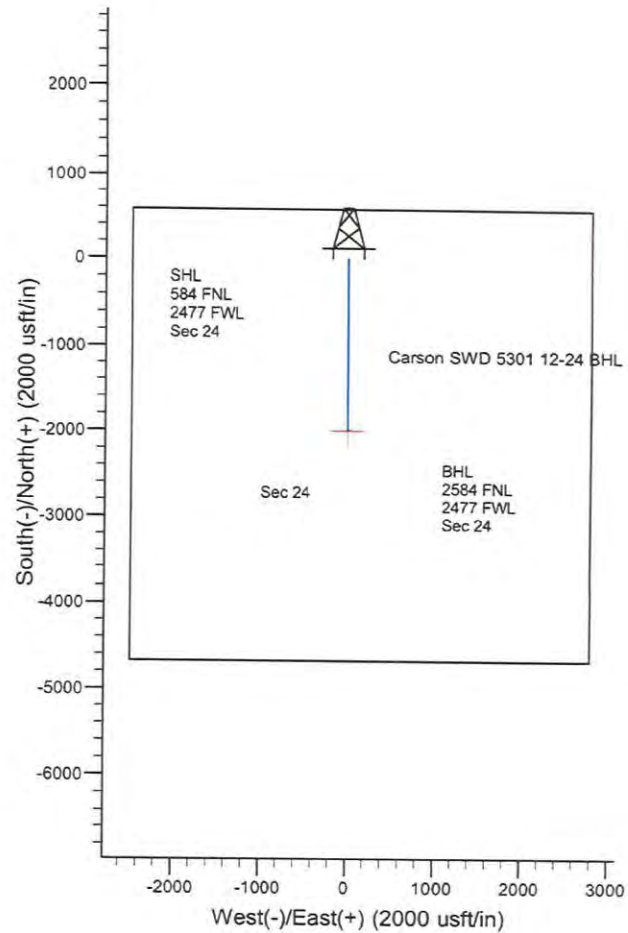
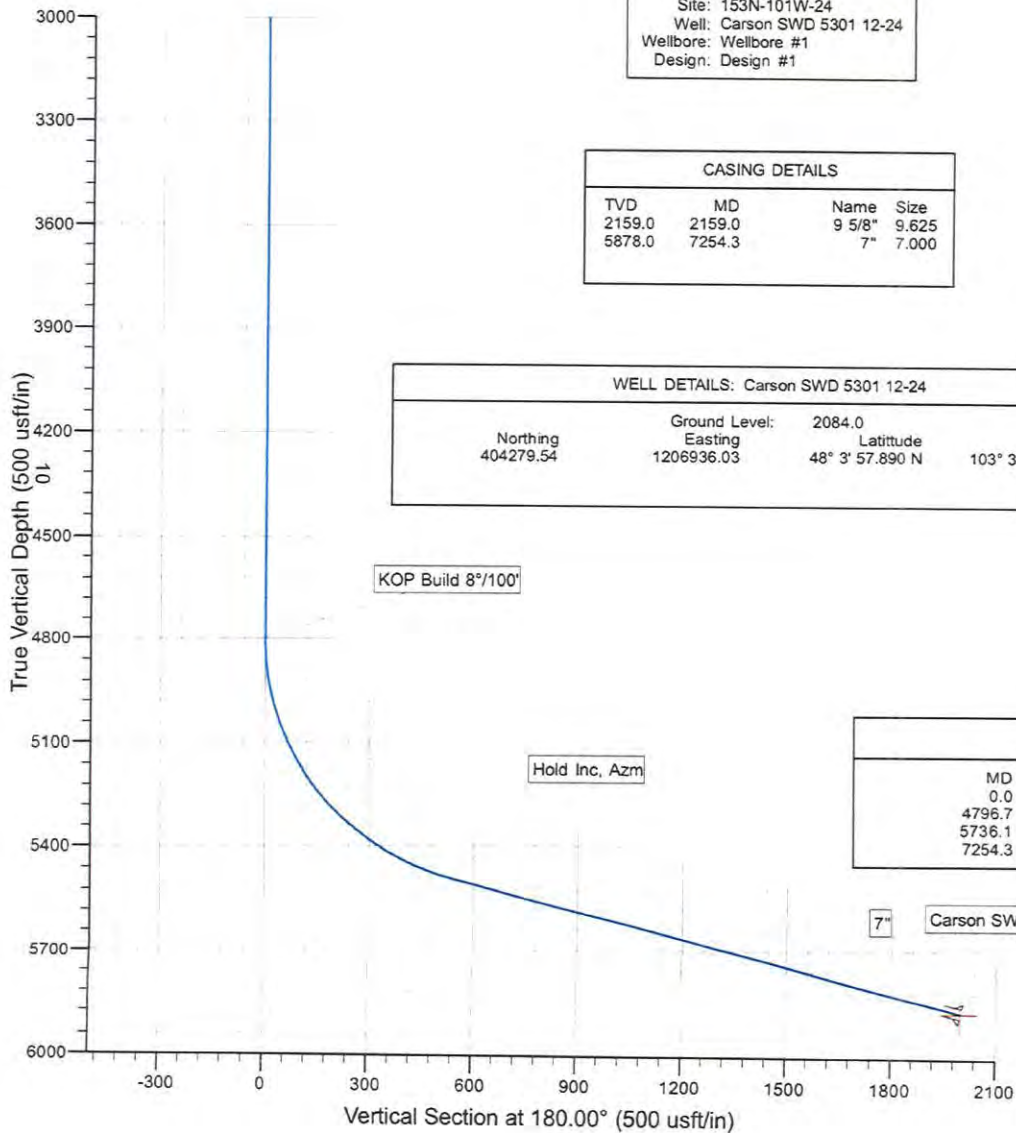
DRILLING PLAN										
PROSPECT/FIELD	Cottonwood Field			Slant Dakota Water Disposal Well	COUNTY/STATE	Mountrail Co., ND				
OPERATOR	Oasis Petroleum			RIG	TBD					
WELL NAME	Carson SWD 5301 12-24									
LOCATION	Surface Location (survey plat):									
EST. T.D.	7,254			GROUND ELEV:		2084 Finished Pad Elev.		Sub Hieght: 25		
PROGNOSIS: Based on 2,109' KB(est)				LOGS: <u>Type</u> <u>Interval</u> OH Logs: None CBL/GR: Above top of cement/GR to base of casing						
<u>MARKER</u>	<u>DEPTH (Surf Loc)</u>	<u>DATUM (Surf Loc)</u>								
Pierre	2,059	50								
Greenhorn	4,606	-2497								
Mowry	4,997	-2888								
Newcastle	5,126	-3017								
Dakota Top	5,421	-3312								
First Dakota Sand Top	5,489	-3380								
First Dakota Sand Base	5,544	-3435								
Second Dakota Sand Top	5,671	-3562								
Second Dakota Sand Base	5,709	-3600								
Third Dakota Sand Top	5,779	-3670								
Third Dakota Sand Base	5,863	-3754								
TD well	5,878	-3769								
Swift	6,061	-3952								
Rierdon	6,377	-4268								
DEVIATION:				8 deg/100' curve slant well						
DST'S:				None planned						
CORES:				None planned						
MUDLOGGING:				30' samples during curve and through Dakota						
BOP:				11" 5000 psi blind, pipe & annular						
Max. Anticipated BHP:				2370						
Surface Formation:				Glacial till						
<u>MUD:</u>	<u>Interval</u>	<u>Type</u>	<u>WT</u>	<u>Vis</u>	<u>WL</u>	<u>Remarks</u>				
Surface	0' -	2,159'	FW/Gel Lime Sweeps	8.6 - 8.9	28-34	NC				
Intermediate	2,159' -	7,254'	Invert	9.0-10.0	40-60	HTHP	Circ Mud Tanks Circ Mud Tanks			
<u>CASING:</u>	<u>Size</u>	<u>Wt ppf</u>	<u>Hole</u>	<u>Depth</u>	<u>Cement</u>	<u>WOC</u>	<u>Remarks</u>			
Surface:	9 5/8"	36	13 1/2"	2,159	To surface	12 hours				
Intermediate:	7"	26	8 3/4"	7,254	2,159	12 hours	Back to surf shoe			
PROBABLE PLUGS, IF REQ'D:										
Plug from TD of vertical = 5,965' to 150' above KOP = 4,800'. Total of 1,140'										
<u>OTHER:</u>	<u>MD</u>	<u>TVD</u>	<u>FNL/FSL</u>	<u>FEL/FWL</u>	<u>S-T-R</u>	<u>ΔZ</u>				
Surface:	2,159	2,159	584' FNL	2477' FWL	T153N-R101W-Sec. 24	180.0	Build Rate: 8 deg /100'			
KOP:	4,797	4,797	584' FNL	2477' FWL	T153N-R101W-Sec. 24	180.0				
EOC:	5,736	5,489	1117' FNL	2477' FWL	T153N-R101W-Sec. 24	180.0				
Dakota Slant TD/ Casing Point	7,254	5,878	2584' FNL	2477' FWL	T153N-R101W-Sec. 24	180.0				
Comments:										
MWD survey every 100'. Place Cement Kickoff Plug										
TIH with 8-3/4" Curve assembly and build curve. MWD Survey every 30'. Land Curve, top of Dakota formation										
Drill slant with 8-3/4" bit and rotate down to TD. MWD Survey every 30'. Run 7" casing and cement										
Reserve pit fluids will be hauled to the nearest disposal site/Reserve pit solids will be solidified and reclaimed in the lined reserve pit										
Geology: BMC 10-30-2014					Engineering: Agonzalez11-25-14					



Project: Indian Hills
Site: 153N-101W-24
Well: Carson SWD 5301 12-24
Wellbore: Wellbore #1
Design: Design #1

CASING DETAILS			
TVD	MD	Name	Size
2159.0	2159.0	9 5/8"	9.625
5878.0	7254.3	7"	7.000

WELL DETAILS: Carson SWD 5301 12-24				
Northing	Ground Level: Easting	2084.0 Latitude	Longitude	
404279.54	1206936.03	48° 3' 57.890 N	103° 36' 56.050 W	



SECTION DETAILS							
MD	Inc	Azi	TVD	+N/-S	+E/-W	Dleg	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	
4796.7	0.00	0.00	4796.7	0.0	0.0	0.00	
5736.1	75.15	180.00	5489.0	-532.7	0.0	8.00	
7254.3	75.15	180.00	5878.0	-2000.2	0.0	0.00	Carson SWD 5301 12-24 BHL

Oasis

Indian Hills

153N-101W-24

Carson SWD 5301 12-24

Wellbore #1

Plan: Design #1

Standard Planning Report

20 March, 2015

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Carson SWD 5301 12-24
Company:	Oasis	TVD Reference:	WELL @ 2109.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2109.0usft (Original Well Elev)
Site:	153N-101W-24	North Reference:	True
Well:	Carson SWD 5301 12-24	Survey Calculation Method:	Minimum Curvature
Wellbore:	Wellbore #1		
Design:	Design #1		

Project	Indian Hills		
Map System:	US State Plane 1983	System Datum:	Mean Sea Level
Geo Datum:	North American Datum 1983		
Map Zone:	North Dakota Northern Zone		

Site	153N-101W-24			
Site Position:		Northing:	404,279.54 usft	Latitude: 48° 3' 57.890 N
From:	Lat/Long	Easting:	1,206,936.03 usft	Longitude: 103° 36' 56.050 W
Position Uncertainty:	0.0 usft	Slot Radius:	13.200 in	Grid Convergence: -2.32 °

Well	Carson SWD 5301 12-24			
Well Position	+N/-S	0.0 usft	Northing:	404,279.54 usft
	+E/-W	0.0 usft	Easting:	1,206,936.03 usft
Position Uncertainty		2.0 usft	Wellhead Elevation:	Ground Level: 2,084.0 usft

Wellbore	Wellbore #1				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	IGRF2010	11/25/2014	8.20	72.96	56,381

Design	Design #1			
Audit Notes:				
Version:	Phase:	PROTOTYPE	Tie On Depth:	0.0
Vertical Section:	Depth From (TVD) (usft)	+N/-S (usft)	+E/-W (usft)	Direction (°)
	0.0	0.0	0.0	180.00

Plan Sections										
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)	TFO (°)	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	0.00	0.00	0.00	
4,796.7	0.00	0.00	4,796.7	0.0	0.0	0.00	0.00	0.00	0.00	
5,736.1	75.15	180.00	5,489.0	-532.7	0.0	8.00	8.00	-19.16	180.00	
7,254.3	75.15	180.00	5,878.0	-2,000.2	0.0	0.00	0.00	0.00	0.00	Carson SWD 5301 12

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Carson SWD 5301 12-24
Company:	Oasis	TVD Reference:	WELL @ 2109.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2109.0usft (Original Well Elev)
Site:	153N-101W-24	North Reference:	True
Well:	Carson SWD 5301 12-24	Survey Calculation Method:	Minimum Curvature
Wellbore:	Wellbore #1		
Design:	Design #1		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
2,160.0	0.00	0.00	2,160.0	0.0	0.0	0.0	0.00	0.00	0.00
2,190.0	0.00	0.00	2,190.0	0.0	0.0	0.0	0.00	0.00	0.00
2,220.0	0.00	0.00	2,220.0	0.0	0.0	0.0	0.00	0.00	0.00
2,250.0	0.00	0.00	2,250.0	0.0	0.0	0.0	0.00	0.00	0.00
2,280.0	0.00	0.00	2,280.0	0.0	0.0	0.0	0.00	0.00	0.00
2,310.0	0.00	0.00	2,310.0	0.0	0.0	0.0	0.00	0.00	0.00
2,340.0	0.00	0.00	2,340.0	0.0	0.0	0.0	0.00	0.00	0.00
2,370.0	0.00	0.00	2,370.0	0.0	0.0	0.0	0.00	0.00	0.00
2,400.0	0.00	0.00	2,400.0	0.0	0.0	0.0	0.00	0.00	0.00
2,430.0	0.00	0.00	2,430.0	0.0	0.0	0.0	0.00	0.00	0.00
2,460.0	0.00	0.00	2,460.0	0.0	0.0	0.0	0.00	0.00	0.00
2,490.0	0.00	0.00	2,490.0	0.0	0.0	0.0	0.00	0.00	0.00
2,520.0	0.00	0.00	2,520.0	0.0	0.0	0.0	0.00	0.00	0.00
2,550.0	0.00	0.00	2,550.0	0.0	0.0	0.0	0.00	0.00	0.00
2,580.0	0.00	0.00	2,580.0	0.0	0.0	0.0	0.00	0.00	0.00
2,610.0	0.00	0.00	2,610.0	0.0	0.0	0.0	0.00	0.00	0.00
2,640.0	0.00	0.00	2,640.0	0.0	0.0	0.0	0.00	0.00	0.00
2,670.0	0.00	0.00	2,670.0	0.0	0.0	0.0	0.00	0.00	0.00
2,700.0	0.00	0.00	2,700.0	0.0	0.0	0.0	0.00	0.00	0.00
2,730.0	0.00	0.00	2,730.0	0.0	0.0	0.0	0.00	0.00	0.00
2,760.0	0.00	0.00	2,760.0	0.0	0.0	0.0	0.00	0.00	0.00
2,790.0	0.00	0.00	2,790.0	0.0	0.0	0.0	0.00	0.00	0.00
2,820.0	0.00	0.00	2,820.0	0.0	0.0	0.0	0.00	0.00	0.00
2,850.0	0.00	0.00	2,850.0	0.0	0.0	0.0	0.00	0.00	0.00
2,880.0	0.00	0.00	2,880.0	0.0	0.0	0.0	0.00	0.00	0.00
2,910.0	0.00	0.00	2,910.0	0.0	0.0	0.0	0.00	0.00	0.00
2,940.0	0.00	0.00	2,940.0	0.0	0.0	0.0	0.00	0.00	0.00
2,970.0	0.00	0.00	2,970.0	0.0	0.0	0.0	0.00	0.00	0.00
3,000.0	0.00	0.00	3,000.0	0.0	0.0	0.0	0.00	0.00	0.00
3,030.0	0.00	0.00	3,030.0	0.0	0.0	0.0	0.00	0.00	0.00
3,060.0	0.00	0.00	3,060.0	0.0	0.0	0.0	0.00	0.00	0.00
3,090.0	0.00	0.00	3,090.0	0.0	0.0	0.0	0.00	0.00	0.00
3,120.0	0.00	0.00	3,120.0	0.0	0.0	0.0	0.00	0.00	0.00
3,150.0	0.00	0.00	3,150.0	0.0	0.0	0.0	0.00	0.00	0.00
3,180.0	0.00	0.00	3,180.0	0.0	0.0	0.0	0.00	0.00	0.00
3,210.0	0.00	0.00	3,210.0	0.0	0.0	0.0	0.00	0.00	0.00
3,240.0	0.00	0.00	3,240.0	0.0	0.0	0.0	0.00	0.00	0.00
3,270.0	0.00	0.00	3,270.0	0.0	0.0	0.0	0.00	0.00	0.00
3,300.0	0.00	0.00	3,300.0	0.0	0.0	0.0	0.00	0.00	0.00
3,330.0	0.00	0.00	3,330.0	0.0	0.0	0.0	0.00	0.00	0.00
3,360.0	0.00	0.00	3,360.0	0.0	0.0	0.0	0.00	0.00	0.00
3,390.0	0.00	0.00	3,390.0	0.0	0.0	0.0	0.00	0.00	0.00
3,420.0	0.00	0.00	3,420.0	0.0	0.0	0.0	0.00	0.00	0.00
3,450.0	0.00	0.00	3,450.0	0.0	0.0	0.0	0.00	0.00	0.00
3,480.0	0.00	0.00	3,480.0	0.0	0.0	0.0	0.00	0.00	0.00
3,510.0	0.00	0.00	3,510.0	0.0	0.0	0.0	0.00	0.00	0.00
3,540.0	0.00	0.00	3,540.0	0.0	0.0	0.0	0.00	0.00	0.00
3,570.0	0.00	0.00	3,570.0	0.0	0.0	0.0	0.00	0.00	0.00
3,600.0	0.00	0.00	3,600.0	0.0	0.0	0.0	0.00	0.00	0.00
3,630.0	0.00	0.00	3,630.0	0.0	0.0	0.0	0.00	0.00	0.00
3,660.0	0.00	0.00	3,660.0	0.0	0.0	0.0	0.00	0.00	0.00
3,690.0	0.00	0.00	3,690.0	0.0	0.0	0.0	0.00	0.00	0.00
3,720.0	0.00	0.00	3,720.0	0.0	0.0	0.0	0.00	0.00	0.00
3,750.0	0.00	0.00	3,750.0	0.0	0.0	0.0	0.00	0.00	0.00
3,780.0	0.00	0.00	3,780.0	0.0	0.0	0.0	0.00	0.00	0.00

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Carson SWD 5301 12-24
Company:	Oasis	TVD Reference:	WELL @ 2109.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2109.0usft (Original Well Elev)
Site:	153N-101W-24	North Reference:	True
Well:	Carson SWD 5301 12-24	Survey Calculation Method:	Minimum Curvature
Wellbore:	Wellbore #1		
Design:	Design #1		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
3,810.0	0.00	0.00	3,810.0	0.0	0.0	0.0	0.00	0.00	0.00
3,840.0	0.00	0.00	3,840.0	0.0	0.0	0.0	0.00	0.00	0.00
3,870.0	0.00	0.00	3,870.0	0.0	0.0	0.0	0.00	0.00	0.00
3,900.0	0.00	0.00	3,900.0	0.0	0.0	0.0	0.00	0.00	0.00
3,930.0	0.00	0.00	3,930.0	0.0	0.0	0.0	0.00	0.00	0.00
3,960.0	0.00	0.00	3,960.0	0.0	0.0	0.0	0.00	0.00	0.00
3,990.0	0.00	0.00	3,990.0	0.0	0.0	0.0	0.00	0.00	0.00
4,020.0	0.00	0.00	4,020.0	0.0	0.0	0.0	0.00	0.00	0.00
4,050.0	0.00	0.00	4,050.0	0.0	0.0	0.0	0.00	0.00	0.00
4,080.0	0.00	0.00	4,080.0	0.0	0.0	0.0	0.00	0.00	0.00
4,110.0	0.00	0.00	4,110.0	0.0	0.0	0.0	0.00	0.00	0.00
4,140.0	0.00	0.00	4,140.0	0.0	0.0	0.0	0.00	0.00	0.00
4,170.0	0.00	0.00	4,170.0	0.0	0.0	0.0	0.00	0.00	0.00
4,200.0	0.00	0.00	4,200.0	0.0	0.0	0.0	0.00	0.00	0.00
4,230.0	0.00	0.00	4,230.0	0.0	0.0	0.0	0.00	0.00	0.00
4,260.0	0.00	0.00	4,260.0	0.0	0.0	0.0	0.00	0.00	0.00
4,290.0	0.00	0.00	4,290.0	0.0	0.0	0.0	0.00	0.00	0.00
4,320.0	0.00	0.00	4,320.0	0.0	0.0	0.0	0.00	0.00	0.00
4,350.0	0.00	0.00	4,350.0	0.0	0.0	0.0	0.00	0.00	0.00
4,380.0	0.00	0.00	4,380.0	0.0	0.0	0.0	0.00	0.00	0.00
4,410.0	0.00	0.00	4,410.0	0.0	0.0	0.0	0.00	0.00	0.00
4,440.0	0.00	0.00	4,440.0	0.0	0.0	0.0	0.00	0.00	0.00
4,470.0	0.00	0.00	4,470.0	0.0	0.0	0.0	0.00	0.00	0.00
4,500.0	0.00	0.00	4,500.0	0.0	0.0	0.0	0.00	0.00	0.00
4,530.0	0.00	0.00	4,530.0	0.0	0.0	0.0	0.00	0.00	0.00
4,560.0	0.00	0.00	4,560.0	0.0	0.0	0.0	0.00	0.00	0.00
4,590.0	0.00	0.00	4,590.0	0.0	0.0	0.0	0.00	0.00	0.00
4,606.0	0.00	0.00	4,606.0	0.0	0.0	0.0	0.00	0.00	0.00
Greenhorn									
4,620.0	0.00	0.00	4,620.0	0.0	0.0	0.0	0.00	0.00	0.00
4,650.0	0.00	0.00	4,650.0	0.0	0.0	0.0	0.00	0.00	0.00
4,680.0	0.00	0.00	4,680.0	0.0	0.0	0.0	0.00	0.00	0.00
4,710.0	0.00	0.00	4,710.0	0.0	0.0	0.0	0.00	0.00	0.00
4,740.0	0.00	0.00	4,740.0	0.0	0.0	0.0	0.00	0.00	0.00
4,770.0	0.00	0.00	4,770.0	0.0	0.0	0.0	0.00	0.00	0.00
4,796.7	0.00	0.00	4,796.7	0.0	0.0	0.0	0.00	0.00	0.00
KOP Build 8°/100'									
4,800.0	0.26	180.00	4,800.0	0.0	0.0	0.0	8.00	8.00	0.00
4,830.0	2.66	180.00	4,830.0	-0.8	0.0	0.8	8.00	8.00	0.00
4,860.0	5.06	180.00	4,859.9	-2.8	0.0	2.8	8.00	8.00	0.00
4,890.0	7.46	180.00	4,889.7	-6.1	0.0	6.1	8.00	8.00	0.00
4,920.0	9.86	180.00	4,919.4	-10.6	0.0	10.6	8.00	8.00	0.00
4,950.0	12.26	180.00	4,948.8	-16.3	0.0	16.3	8.00	8.00	0.00
4,980.0	14.66	180.00	4,978.0	-23.3	0.0	23.3	8.00	8.00	0.00
4,999.7	16.24	180.00	4,997.0	-28.6	0.0	28.6	8.00	8.00	0.00
Mowry									
5,010.0	17.06	180.00	5,006.9	-31.5	0.0	31.5	8.00	8.00	0.00
5,040.0	19.46	180.00	5,035.3	-40.9	0.0	40.9	8.00	8.00	0.00
5,070.0	21.86	180.00	5,063.4	-51.5	0.0	51.5	8.00	8.00	0.00
5,100.0	24.26	180.00	5,091.0	-63.3	0.0	63.3	8.00	8.00	0.00
5,130.0	26.66	180.00	5,118.1	-76.2	0.0	76.2	8.00	8.00	0.00
5,160.0	29.06	180.00	5,144.6	-90.2	0.0	90.2	8.00	8.00	0.00
5,190.0	31.46	180.00	5,170.5	-105.3	0.0	105.3	8.00	8.00	0.00
5,220.0	33.86	180.00	5,195.8	-121.5	0.0	121.5	8.00	8.00	0.00

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Carson SWD 5301 12-24
Company:	Oasis	TVD Reference:	WELL @ 2109.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2109.0usft (Original Well Elev)
Site:	153N-101W-24	North Reference:	True
Well:	Carson SWD 5301 12-24	Survey Calculation Method:	Minimum Curvature
Wellbore:	Wellbore #1		
Design:	Design #1		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
5,250.0	36.26	180.00	5,220.3	-138.7	0.0	138.7	8.00	8.00	0.00
5,280.0	38.66	180.00	5,244.1	-157.0	0.0	157.0	8.00	8.00	0.00
5,310.0	41.06	180.00	5,267.2	-176.2	0.0	176.2	8.00	8.00	0.00
5,340.0	43.46	180.00	5,289.4	-196.4	0.0	196.4	8.00	8.00	0.00
5,370.0	45.86	180.00	5,310.7	-217.5	0.0	217.5	8.00	8.00	0.00
5,400.0	48.26	180.00	5,331.1	-239.4	0.0	239.4	8.00	8.00	0.00
5,430.0	50.66	180.00	5,350.6	-262.2	0.0	262.2	8.00	8.00	0.00
5,460.0	53.06	180.00	5,369.2	-285.8	0.0	285.8	8.00	8.00	0.00
5,490.0	55.46	180.00	5,386.7	-310.2	0.0	310.2	8.00	8.00	0.00
5,520.0	57.86	180.00	5,403.2	-335.2	0.0	335.2	8.00	8.00	0.00
5,550.0	60.26	180.00	5,418.6	-361.0	0.0	361.0	8.00	8.00	0.00
5,554.9	60.66	180.00	5,421.0	-365.2	0.0	365.2	8.00	8.00	0.00
Dakota Top									
5,580.0	62.66	180.00	5,432.9	-387.3	0.0	387.3	8.00	8.00	0.00
5,610.0	65.06	180.00	5,446.1	-414.2	0.0	414.2	8.00	8.00	0.00
5,640.0	67.46	180.00	5,458.2	-441.7	0.0	441.7	8.00	8.00	0.00
5,670.0	69.86	180.00	5,469.1	-469.6	0.0	469.6	8.00	8.00	0.00
5,700.0	72.26	180.00	5,478.9	-498.0	0.0	498.0	8.00	8.00	0.00
5,730.0	74.66	180.00	5,487.4	-526.8	0.0	526.8	8.00	8.00	0.00
5,736.1	75.15	180.00	5,489.0	-532.7	0.0	532.7	8.00	8.00	0.00
Hold Inc, Azm									
5,736.2	75.15	180.00	5,489.0	-532.7	0.0	532.7	1.57	1.57	0.00
First Dakota Sand Top									
5,760.0	75.15	180.00	5,495.1	-555.8	0.0	555.8	0.00	0.00	0.00
5,790.0	75.15	180.00	5,502.8	-584.8	0.0	584.8	0.00	0.00	0.00
5,820.0	75.15	180.00	5,510.5	-613.8	0.0	613.8	0.00	0.00	0.00
5,850.0	75.15	180.00	5,518.2	-642.8	0.0	642.8	0.00	0.00	0.00
5,880.0	75.15	180.00	5,525.9	-671.8	0.0	671.8	0.00	0.00	0.00
5,910.0	75.15	180.00	5,533.5	-700.8	0.0	700.8	0.00	0.00	0.00
5,940.0	75.15	180.00	5,541.2	-729.8	0.0	729.8	0.00	0.00	0.00
5,950.8	75.15	180.00	5,544.0	-740.2	0.0	740.2	0.00	0.00	0.00
First Dakota Sand Base									
5,970.0	75.15	180.00	5,548.9	-758.8	0.0	758.8	0.00	0.00	0.00
6,000.0	75.15	180.00	5,556.6	-787.8	0.0	787.8	0.00	0.00	0.00
6,030.0	75.15	180.00	5,564.3	-816.8	0.0	816.8	0.00	0.00	0.00
6,060.0	75.15	180.00	5,572.0	-845.8	0.0	845.8	0.00	0.00	0.00
6,090.0	75.15	180.00	5,579.7	-874.8	0.0	874.8	0.00	0.00	0.00
6,120.0	75.15	180.00	5,587.4	-903.8	0.0	903.8	0.00	0.00	0.00
6,150.0	75.15	180.00	5,595.0	-932.8	0.0	932.8	0.00	0.00	0.00
6,180.0	75.15	180.00	5,602.7	-961.7	0.0	961.7	0.00	0.00	0.00
6,210.0	75.15	180.00	5,610.4	-990.7	0.0	990.7	0.00	0.00	0.00
6,240.0	75.15	180.00	5,618.1	-1,019.7	0.0	1,019.7	0.00	0.00	0.00
6,270.0	75.15	180.00	5,625.8	-1,048.7	0.0	1,048.7	0.00	0.00	0.00
6,300.0	75.15	180.00	5,633.5	-1,077.7	0.0	1,077.7	0.00	0.00	0.00
6,330.0	75.15	180.00	5,641.2	-1,106.7	0.0	1,106.7	0.00	0.00	0.00
6,360.0	75.15	180.00	5,648.8	-1,135.7	0.0	1,135.7	0.00	0.00	0.00
6,390.0	75.15	180.00	5,656.5	-1,164.7	0.0	1,164.7	0.00	0.00	0.00
6,420.0	75.15	180.00	5,664.2	-1,193.7	0.0	1,193.7	0.00	0.00	0.00
6,446.5	75.15	180.00	5,671.0	-1,219.3	0.0	1,219.3	0.00	0.00	0.00
Second Dakota Sand Top									
6,450.0	75.15	180.00	5,671.9	-1,222.7	0.0	1,222.7	0.00	0.00	0.00
6,480.0	75.15	180.00	5,679.6	-1,251.7	0.0	1,251.7	0.00	0.00	0.00
6,510.0	75.15	180.00	5,687.3	-1,280.7	0.0	1,280.7	0.00	0.00	0.00
6,540.0	75.15	180.00	5,695.0	-1,309.7	0.0	1,309.7	0.00	0.00	0.00
6,570.0	75.15	180.00	5,702.7	-1,338.7	0.0	1,338.7	0.00	0.00	0.00

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Carson SWD 5301 12-24
Company:	Oasis	TVD Reference:	WELL @ 2109.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2109.0usft (Original Well Elev)
Site:	153N-101W-24	North Reference:	True
Well:	Carson SWD 5301 12-24	Survey Calculation Method:	Minimum Curvature
Wellbore:	Wellbore #1		
Design:	Design #1		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
6,594.8	75.15	180.00	5,709.0	-1,362.7	0.0	1,362.7	0.00	0.00	0.00
Second Dakota Sand Base									
6,600.0	75.15	180.00	5,710.3	-1,367.7	0.0	1,367.7	0.00	0.00	0.00
6,630.0	75.15	180.00	5,718.0	-1,396.7	0.0	1,396.7	0.00	0.00	0.00
6,660.0	75.15	180.00	5,725.7	-1,425.7	0.0	1,425.7	0.00	0.00	0.00
6,690.0	75.15	180.00	5,733.4	-1,454.7	0.0	1,454.7	0.00	0.00	0.00
6,720.0	75.15	180.00	5,741.1	-1,483.7	0.0	1,483.7	0.00	0.00	0.00
6,750.0	75.15	180.00	5,748.8	-1,512.7	0.0	1,512.7	0.00	0.00	0.00
6,780.0	75.15	180.00	5,756.5	-1,541.7	0.0	1,541.7	0.00	0.00	0.00
6,810.0	75.15	180.00	5,764.2	-1,570.7	0.0	1,570.7	0.00	0.00	0.00
6,840.0	75.15	180.00	5,771.8	-1,599.7	0.0	1,599.7	0.00	0.00	0.00
6,867.9	75.15	180.00	5,779.0	-1,626.7	0.0	1,626.7	0.00	0.00	0.00
Third Dakota Sand Top									
6,870.0	75.15	180.00	5,779.5	-1,628.7	0.0	1,628.7	0.00	0.00	0.00
6,900.0	75.15	180.00	5,787.2	-1,657.7	0.0	1,657.7	0.00	0.00	0.00
6,930.0	75.15	180.00	5,794.9	-1,686.7	0.0	1,686.7	0.00	0.00	0.00
6,960.0	75.15	180.00	5,802.6	-1,715.7	0.0	1,715.7	0.00	0.00	0.00
6,990.0	75.15	180.00	5,810.3	-1,744.7	0.0	1,744.7	0.00	0.00	0.00
7,020.0	75.15	180.00	5,818.0	-1,773.7	0.0	1,773.7	0.00	0.00	0.00
7,050.0	75.15	180.00	5,825.6	-1,802.7	0.0	1,802.7	0.00	0.00	0.00
7,080.0	75.15	180.00	5,833.3	-1,831.7	0.0	1,831.7	0.00	0.00	0.00
7,110.0	75.15	180.00	5,841.0	-1,860.7	0.0	1,860.7	0.00	0.00	0.00
7,140.0	75.15	180.00	5,848.7	-1,889.7	0.0	1,889.7	0.00	0.00	0.00
7,170.0	75.15	180.00	5,856.4	-1,918.7	0.0	1,918.7	0.00	0.00	0.00
7,195.8	75.15	180.00	5,863.0	-1,943.6	0.0	1,943.6	0.00	0.00	0.00
Third Dakota Sand Base									
7,200.0	75.15	180.00	5,864.1	-1,947.7	0.0	1,947.7	0.00	0.00	0.00
7,230.0	75.15	180.00	5,871.8	-1,976.7	0.0	1,976.7	0.00	0.00	0.00
7,254.3	75.15	180.00	5,878.0	-2,000.2	0.0	2,000.2	0.00	0.00	0.00
TD well - 7"									

Design Targets									
Target Name	Dip Angle (°)	Dip Dir. (°)	TVD (usft)	+N/-S (usft)	+E/-W (usft)	Northing (usft)	Easting (usft)	Latitude	Longitude
- hit/miss target									
- Shape									
Carson SWD 5301 12-24	0.00	0.00	5,878.0	-2,000.2	0.0	402,280.98	1,206,855.12	48° 3' 38.150 N	103° 36' 56.050 W
- plan hits target center									
- Point									

Casing Points					
Measured Depth (usft)	Vertical Depth (usft)	Name	Casing Diameter (in)	Hole Diameter (in)	
2,159.0	2,159.0	9 5/8"	9.625	12.250	
7,254.3	5,878.0	7"	7.000	8.750	

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Carson SWD 5301 12-24
Company:	Oasis	TVD Reference:	WELL @ 2109.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2109.0usft (Original Well Elev)
Site:	153N-101W-24	North Reference:	True
Well:	Carson SWD 5301 12-24	Survey Calculation Method:	Minimum Curvature
Wellbore:	Wellbore #1		
Design:	Design #1		

Formations						
Measured Depth (usft)	Vertical Depth (usft)	Name	Lithology	Dip (°)	Dip Direction (°)	
2,059.0	2,152.0	Pierre				
4,606.0	4,699.0	Greenhorn				
4,999.7	5,090.0	Mowry				
5,554.9	5,514.0	Dakota Top				
5,736.2	5,582.0	First Dakota Sand Top				
5,950.8	5,637.0	First Dakota Sand Base				
6,446.5	5,764.0	Second Dakota Sand Top				
6,594.8	5,802.0	Second Dakota Sand Base				
6,867.9	5,872.0	Third Dakota Sand Top				
7,195.8	5,956.0	Third Dakota Sand Base				
7,254.3	5,971.0	TD well				

Plan Annotations					
Measured Depth (usft)	Vertical Depth (usft)	Local Coordinates			
		+N/-S (usft)	+E/-W (usft)	Comment	
4,796.7	4,796.7	0.0	0.0	KOP Build 8°/100'	
5,736.1	5,489.0	-532.7	0.0	Hold Inc, Azm	

**Oasis Petroleum
Well Summary
Carson SWD 5301 12-24
Section 24 T153N R101W
Mckenzie County, ND**

SURFACE CASING AND CEMENT DESIGN

							Make-up Torque (ft-lbs)		
Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Minimum	Optimum	Max
9-5/8"	0' - 2159'	36	J-55	LTC	8.921"	8.765"	3400	4530	5660

Interval	Description	Collapse	Burst	Tension
		(psi) / a	(psi) / b	(1000 lbs) / c
0' - 2159'	9-5/8", 36#, J-55, LTC, 8rd	2020 / 1.99	3520 / 3.48	453 / 2.71

API Rating & Safety Factor

- a) Collapse based on full casing evacuation with 9 ppg fluid on backside (2159' setting)
- b) Burst pressure based on 9 ppg fluid with no fluid on backside (2159' setting depth).
- c) Based on string weight in 9 ppg fluid at 2159' TVD plus 100k# overpull. (Buoyed weight equals 67k lbs.)

Cement volumes are based on 9-5/8" casing set in 13-1/2 " hole with 60% excess to circulate cement back to surface.
Mix and pump the following slurry.

Pre-flush (Spacer): **20 bbls** fresh water

Lead Slurry: **462 sks** (244 bbls), 11.5 lb/gal, 2.97 cu. Ft./sk Varicem Cement with 0.125 il/sk Lost Circulation Additive

Tail Slurry: **173 sks** (62 bbls), 13.0 lb/gal, 2.01 cu.ft./sk Varicem with .125 lb/sk Lost Circulation Agent

**Oasis Petroleum
Well Summary
Carson SWD 5301 12-24
Section 24 T153N R101W
Mckenzie County, ND**

INTERMEDIATE CASING AND CEMENT DESIGN

Size	Interval	Weight	Grade	Coupling	I.D.	Drift**	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
7"	0' - 7254'	26	L-80	LTC	6.276"	6.151	3900	4900	5900

**Special Drift"

Interval	Length	Description	Collapse	Burst	Tension
			(psi) a	(psi) b	(1000 lbs) c
0' - 7254'	7254'	7", 26#, L-80, LTC, 8rd	5410 / 1.76*	7240 / 1.44	519 / 1.99

API Rating & Safety Factor

- a) *Assume full casing evacuation with 10 ppg fluid on backside. **Assume full casing evacuation with psi/ft equivalent fluid gradient across salt intervals.
- b) Burst pressure based on 5000 psig max press for stimulation plus 10 ppg fluid in casing and 10 ppg fluid on backside-to 5878' TVD.
- c) Based on string weight in 10 ppg fluid, (160k lbs buoyed weight) plus 100k lbs overpull.

Cement volumes are estimates based on 7" casing set in an 8-3/4" hole with 30% excess.

Mix and pump the following slurry

Pre-flush (Spacer): **100 bbls** Saltwater
 20bbls Tuned Spacer III

Lead Slurry: **216 sks** (100 bbls), 11.8 ppg, 2.55 cu. ft./sk Econocem Cement with .3% Fe-2 and .25 lb/sk Lost Circulation Additive

Tail Slurry: **293 sks** (86 bbls), 14.0 ppg, 1.55 cu. ft./sk Extendcem System with .2% HR-5 Retarder and .25 lb/sk Lost Circulation Additive



12/9/2014

Victoria Siemieniewski
Regulatory Specialist
Oasis Petroleum
1001 Fannin St.
Suite 1500
Houston, TX 77002

Kevin Connors
UIC/CCS Supervisor
North Dakota Industrial Commission
600 East Boulevard Avenue Dept. 405
Bismarck, ND 58505-0840

RE: Carson SWD 5301 12-24
Request for a legal street address

Dear Mr. Connors:

Oasis Petroleum has requested a physical street address for the subject well/well facility. The request was made to Aaron Chisholm, GIS Specialist, McKenzie County. Upon receiving a legal street address, Oasis will submit the address to the NDIC on a Sundry Notice (form 4) pursuant to 43-02-03-28.

Thank you for your consideration.

Respectfully,

A handwritten signature in black ink, appearing to read "Victoria Siemieniewski".

Victoria Siemieniewski
Regulatory Specialist
Oasis Petroleum
281-404-9652



3/3/2015

Michael Kukuk
Regulatory Supervisor
Oasis Petroleum
1001 Fannin St.
Suite 1500
Houston, TX 77002

Kevin Connors
UIC/CCS Supervisor
North Dakota Industrial Commission
600 East Boulevard Avenue Dept. 405
Bismarck, ND 58505-0840

RE: Carson SWD 5301 12-24
Request for approach permit

Dear Mr. Connors:

Oasis Petroleum has requested a county approach permit for the subject well. If you prefer, Oasis will submit confirmation of receipt to your attention via email.

Thank you for your consideration.

Respectfully,

A handwritten signature in black ink, appearing to read "Michael Kukuk", written in a cursive style.

Michael Kukuk
Regulatory Supervisor
Oasis Petroleum
281-404-9575